

MASTER : PROJECT REPORT

CENTRALESUPELEC-UNIVERSITE PARIS SACLAY
PARCOURS : CONTROL ENGINEERING

Quantum control : Linear observer for parameter phase estimation

Student: Boris-Alexandre Baudel

Location: Laboratoire des Signaux et Systèmes (L2S), CentraleSupélec

Department: Modélisation pour la commande des Systèmes Dynamiques

Supervisors: Prof. Nina Amini, Dr. Rebbecca TY Thien

Email : boris.baudel@student-cs.fr



CentraleSupélec



université
PARIS-SACLAY

Chapitre 1 – Introduction

1. Motivation

Optical phase carries information that classical intensity measurements cannot access: the relative time delay between the arms of an interferometer, the displacement of an optomechanical mirror, the gravitational deformation of spacetime in LIGO-class detectors [1], and the quantum signal of continuous-variable quantum key distribution [2]. In all these settings the photodetector never measures phase directly; the phase must instead be inferred from a noisy quadrature record produced by balanced homodyne detection [3, 4, 5]. Estimating an unknown optical phase from noisy continuous measurements is therefore a hidden-state estimation problem at the intersection of quantum optics, control theory and stochastic estimation. It has motivated decades of work, from Belavkin’s foundational quantum filtering [6, 7], to Wiseman’s adaptive homodyne strategies [8, 9, 10, 11], to coherent quantum observers [12, 13, 14, 15, 16], and to modern continuous parameter estimation in optomechanics and metrology [17, 18, 19, 20, 21, 22]. The practical question that ties all this work together is the same:

Given a noisy classical measurement record, how accurately can we reconstruct a hidden phase that affects the cavity field only indirectly?

2. Problem statement

This report addresses the question above in three increasingly realistic formulations. (i) In a scalar formulation, the hidden phase $\varphi(t)$ is modeled as an Ornstein–Uhlenbeck process and observed directly through a noisy homodyne record after local linearization, yielding a classical Kalman–Bucy benchmark. (ii) In a single-cavity formulation, the phase enters the coherent drive of an optical cavity and the homodyne detector observes only one quadrature of the output field; the phase is no longer directly visible and must first perturb the intracavity quadratures, raising the structural question under what conditions is the phase direction observable through the homodyne channel? (iii) In a two-cavity formulation, an auxiliary cavity is coupled to the sensing cavity before measurement, either through a Hamiltonian (bidirectional) coupling typical of linear quantum networks [23, 24, 25], or through a unidirectional cascade coupling in the spirit of Gardiner’s input–output formalism [26]. The auxiliary cavity does not observe $\varphi(t)$ directly; it only reshapes the dynamical pathway through which phase information reaches the detector. The central question becomes: *when does this preprocessing actually improve estimation, and when does it instead inject additional noise that degrades it?*

3. Contributions

1. **Finite-horizon directional observability metric.** We define $J_\varphi(T) := e_\varphi^\top W_o(T) e_\varphi$ as the projection of the observability Gramian onto the phase direction, prove it is exactly the deterministic output energy generated by a unit perturbation along that direction, and obtain a directional Cramér–Rao lower bound on phase variance over $[0, T]$.
2. **Closed-form steady-state phase-variance decomposition.** For the resonant single-cavity model, we prove that the stationary phase-error variance equals the open-loop Ornstein–Uhlenbeck variance minus a measurement-induced correction explicitly proportional to the closed-loop covariance between the measured quadrature and the hidden phase. This explains why $J_\varphi(T)$ is a faithful intra-family predictor but cannot collapse the root-mean-square error to a single universal curve across stochastic regimes.
3. **Visibility/noise trade-off in augmented preprocessing.** A Hamiltonian-coupled two-cavity model is studied through a relative visibility gain ΔJ_φ and a relative noise burden ΔQ ; we map numerically the operating regimes (homodyne angle, OU rate, phase coupling, linewidth) in which the auxiliary cavity actually improves estimation, finding it beneficial in geometry-dominated regimes and detrimental in noise-dominated ones.
4. **Adaptive EKF for the nonlinear cascade architecture.** On the unidirectional cascade two-cavity model with full trigonometric nonlinearity, we implement an extended Kalman filter combined with an adaptive local-oscillator phase $\theta_k = \text{wrap}(\hat{\varphi}_{k|k-1} + \pi/2)$; on a representative trajectory it outperforms the best fixed-angle baseline by $\approx 20\%$ in tail-RMSE, with no fringe-hopping in the small-amplitude regime considered.

4. Outline of the report

Chapter 2 introduces balanced homodyne detection and the linearized state-space measurement model. Chapter 2 develops the scalar Kalman–Bucy benchmark and contrasts it with a deterministic Luenberger observer. Chapter 3 builds the open-cavity model under the input–output Markov approximation, linearizes around the coherent operating point, and assembles the augmented state-space model on which the analysis is based. Chapter 4 contains the structural results: $J_\varphi(T)$, the resonant phase-visibility condition, the output-energy interpretation, the directional Cramér–Rao bound, the closed form of $J_\varphi(\infty)$, the steady-state variance decomposition, and the Hamiltonian-coupled two-cavity trade-off study. Chapter 5 drops the small-phase approximation and benchmarks the adaptive EKF against fixed-angle baselines on the unidirectional cascade model. Chapter 8 collects the detailed algebraic derivations and the complete table of symbols.

Chapitre 2 – Homodyne Detection

5. Context and role of homodyne detection in the project

The general motivation, problem statement, and contributions of this report are given in the Introduction. The purpose of the present chapter is more focused: to introduce balanced homodyne detection, which is the measurement principle used throughout the report, and to show how an optical phase difference is converted into a classical stochastic signal that can be fed to the state-space estimators of the next chapters. Throughout this report, time is denoted by t , and the physical quantity to be estimated is an optical phase

$$\varphi(t), \quad (1)$$

a real-valued time-dependent variable representing the unknown phase fluctuation carried by an optical field. The phase $\varphi(t)$ is not assumed to be directly measurable: the detector outputs only a noisy optical quadrature whose value depends, directly or indirectly, on $\varphi(t)$. The central estimation problem of the report is therefore to reconstruct $\varphi(t)$ from this noisy quadrature record.

6. Balanced homodyne detection

Balanced homodyne detection measures one quadrature of an optical field. The optical field of interest is mixed with a strong coherent reference field, called the local oscillator, on a balanced 50/50 beam splitter. The two output fields are sent to two photodiodes. The two photocurrents are then subtracted. This subtraction cancels the large common optical intensity and keeps the phase-sensitive interference term between the signal and the local oscillator [3, 4, 5]. The field carrying the unknown phase is called the signal field. Its annihilation operator is denoted by

$$\hat{a}_s(t),$$

where the hat indicates a quantum operator and the subscript s stands for signal. When the signal field is in a coherent state, its mean complex amplitude is written as

$$a_s(t) = A_s e^{i\varphi(t)}. \quad (2)$$

In this expression, $a_s(t)$ is the classical coherent amplitude associated with the quantum field $\hat{a}_s(t)$, $A_s > 0$ is the real signal amplitude, i is the imaginary unit, and $\varphi(t)$ is the optical phase to be estimated. The reference field is the local oscillator. Its annihilation operator is denoted by

$$\hat{b}_{\text{LO}}(t),$$

where the subscript LO stands for local oscillator. Its coherent amplitude is written as

$$b_{\text{LO}}(t) = B_{\text{LO}} e^{i\theta(t)}. \quad (3)$$

Here, $b_{\text{LO}}(t)$ is the classical coherent amplitude of the local oscillator, $B_{\text{LO}} > 0$ is its real amplitude, and $\theta(t)$ is its controllable optical phase. In homodyne detection, the local oscillator is chosen much stronger than the signal field:

$$B_{\text{LO}} \gg A_s. \quad (4)$$

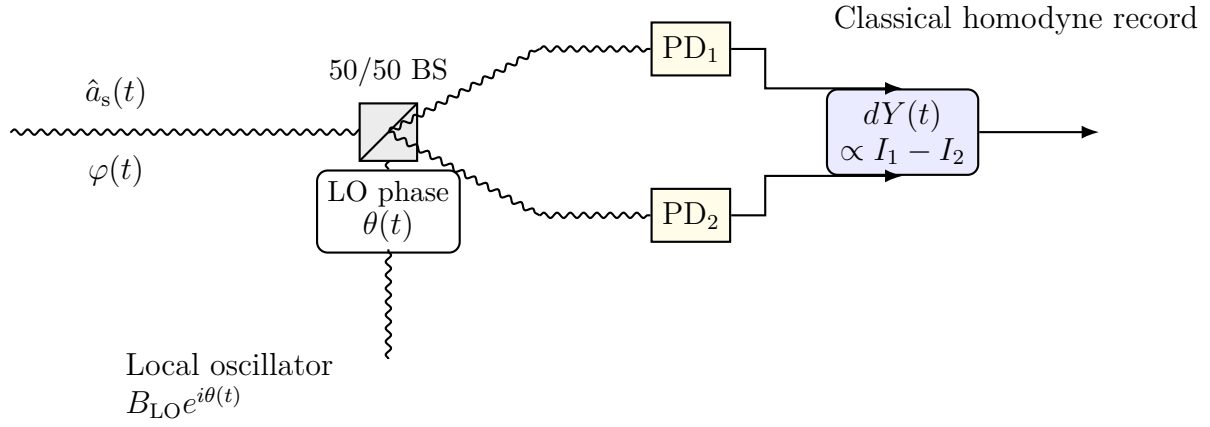


Figure 1: Balanced homodyne detection. The signal field carrying the unknown phase $\varphi(t)$ is mixed with a strong local oscillator of phase $\theta(t)$. The difference between the two photodiode currents removes the large common intensity contribution and produces a phase-sensitive classical measurement record.

This condition means that the local oscillator can be treated as a strong classical phase reference. The phase $\theta(t)$ determines the quadrature direction selected by the measurement.

The signal field and the local oscillator are mixed on a lossless balanced beam splitter. In this chapter, the beam-splitter output fields are denoted by $\hat{c}(t)$ and $\hat{d}(t)$. The convention used in the main text is

$$\hat{c}(t) = \frac{\hat{a}_s(t) + \hat{b}_{LO}(t)}{\sqrt{2}}, \quad \hat{d}(t) = \frac{\hat{a}_s(t) - \hat{b}_{LO}(t)}{\sqrt{2}}. \quad (5)$$

The factor $1/\sqrt{2}$ expresses the fact that the beam splitter is balanced: each output contains one half of the optical intensity, up to the chosen phase convention. The output field $\hat{c}(t)$ is sent to the first photodiode, and the output field $\hat{d}(t)$ is sent to the second photodiode. A photodiode measures an intensity proportional to a photon number. The corresponding photocurrents are therefore modeled as

$$\hat{I}_1(t) \propto \hat{c}^\dagger(t)\hat{c}(t), \quad \hat{I}_2(t) \propto \hat{d}^\dagger(t)\hat{d}(t), \quad (6)$$

where $(\cdot)^\dagger$ denotes the adjoint operator. The balanced homodyne signal is the difference between the two photocurrents,

$$\hat{I}_1(t) - \hat{I}_2(t).$$

Using (5), the difference current is

$$\begin{aligned} \hat{c}^\dagger\hat{c} - \hat{d}^\dagger\hat{d} &= \frac{1}{2} \left[(\hat{a}_s^\dagger + \hat{b}_{LO}^\dagger) (\hat{a}_s + \hat{b}_{LO}) - (\hat{a}_s^\dagger - \hat{b}_{LO}^\dagger) (\hat{a}_s - \hat{b}_{LO}) \right] \\ &= \hat{a}_s^\dagger\hat{b}_{LO} + \hat{b}_{LO}^\dagger\hat{a}_s. \end{aligned} \quad (7)$$

The direct signal intensity $\hat{a}_s^\dagger\hat{a}_s$ and the direct local-oscillator intensity $\hat{b}_{LO}^\dagger\hat{b}_{LO}$ cancel. The remaining term is the interference between the signal field and the local oscillator. This is the term that carries information about the phase difference between the two fields. Since the local oscillator is strong, it is treated as a classical coherent amplitude:

$$\hat{b}_{LO}(t) \simeq B_{LO}e^{i\theta(t)}. \quad (8)$$

Substitution into (7) gives

$$\hat{c}^\dagger\hat{c} - \hat{d}^\dagger\hat{d} \simeq B_{LO} \left(e^{i\theta(t)}\hat{a}_s^\dagger(t) + e^{-i\theta(t)}\hat{a}_s(t) \right). \quad (9)$$

The expression in parentheses is the field quadrature selected by the local oscillator phase $\theta(t)$.

If the signal field is also replaced by its coherent amplitude $a_s(t) = A_s e^{i\varphi(t)}$, then the classical difference current satisfies

$$\begin{aligned} I_1(t) - I_2(t) &\propto a_s^*(t)b_{\text{LO}}(t) + b_{\text{LO}}^*(t)a_s(t) \\ &= 2A_s B_{\text{LO}} \cos(\varphi(t) - \theta(t)). \end{aligned} \quad (10)$$

Here, $(\cdot)^*$ denotes complex conjugation. The current is therefore sensitive to the phase difference $\varphi(t) - \theta(t)$. This is the basic physical mechanism by which homodyne detection converts optical phase information into a measurable electrical signal. After calibration by the known local-oscillator amplitude and detector gain, the homodyne record is modeled as the stochastic differential equation

$$dY(t) = K_h \cos(\varphi(t) - \theta(t)) dt + \sigma_h dV(t). \quad (11)$$

In this equation, $Y(t)$ is the integrated classical homodyne record, $dY(t)$ is the measurement increment over the infinitesimal time interval dt , $K_h > 0$ is the calibrated homodyne gain, and $\sigma_h > 0$ is the measurement-noise amplitude. The process $V(t)$ is a standard Wiener process, and $dV(t)$ represents the shot-noise increment of the homodyne measurement. In ideal normalized units, the homodyne gain is often written as

$$K_h = 2\sqrt{\eta} A_s, \quad 0 \leq \eta \leq 1, \quad (12)$$

where η is the detection efficiency. The exact numerical prefactor depends on the quadrature normalization and detector calibration conventions [3, 4, 5].

7. Linearized phase measurement

The nonlinear measurement model (11) shows that the homodyne signal depends on $\cos(\varphi(t) - \theta(t))$. For Kalman filtering and linear observer design, it is useful to obtain a measurement equation that is locally linear in the unknown phase. This is done by choosing an operating point and linearizing the homodyne current around that point. Let φ_0 be a nominal phase. The phase error is defined by

$$\delta\varphi(t) := \varphi(t) - \varphi_0. \quad (13)$$

The variable $\delta\varphi(t)$ measures the deviation of the true phase from the nominal phase. The linear approximation is valid when $|\delta\varphi(t)| \ll 1$. The local oscillator is chosen in phase quadrature with respect to the nominal signal:

$$\theta(t) = \varphi_0 + \frac{\pi}{2}. \quad (14)$$

With this choice,

$$\begin{aligned} \cos(\varphi(t) - \theta(t)) &= \cos\left(\varphi_0 + \delta\varphi(t) - \varphi_0 - \frac{\pi}{2}\right) \\ &= \cos\left(\delta\varphi(t) - \frac{\pi}{2}\right) \\ &= \sin(\delta\varphi(t)). \end{aligned} \quad (15)$$

For small phase errors, Taylor expansion gives

$$\sin(\delta\varphi(t)) = \delta\varphi(t) + \mathcal{O}(\delta\varphi(t)^3). \quad (16)$$

Keeping only the first-order term, the nonlinear homodyne model becomes

$$dY(t) \simeq K_h \delta\varphi(t) dt + \sigma_h dV(t). \quad (17)$$

This is a local linear measurement equation. It says that the measurement increment is approximately proportional to the small phase error, plus measurement noise. It is convenient to normalize the measurement by the gain K_h . We define the normalized homodyne record $y_c(t)$ by dividing the raw record by K_h . Then (17) becomes

$$dy_c(t) = \delta\varphi(t) dt + \sqrt{R_h} dV(t), \quad R_h = \frac{\sigma_h^2}{K_h^2}. \quad (18)$$

Here, $R_h > 0$ is the effective measurement-noise intensity after normalization. If the nominal phase is chosen as $\varphi_0 = 0$, then $\delta\varphi(t) = \varphi(t)$. In ideal units with $\eta = 1$, $\sigma_h = 1$, and $K_h = 2A_s$, the normalized measurement becomes

$$dy_c(t) = \varphi(t) dt + \frac{1}{2A_s} dV(t). \quad (19)$$

Using the formal white-noise notation

$$w(t) = \frac{dV(t)}{dt},$$

the same model is often written as

$$y_c(t) = \varphi(t) + \frac{1}{2A_s} w(t). \quad (20)$$

This equation is a compact engineering notation. The mathematically precise form is the stochastic differential equation (19). The term $w(t)$ represents ideal white measurement noise. The coefficient $1/(2A_s)$ shows that increasing the signal amplitude improves the effective signal-to-noise ratio. In the first estimation model used later in the report, the hidden phase is modeled as an Ornstein–Uhlenbeck process:

$$\boxed{d\varphi(t) = -\lambda\varphi(t) dt + \sqrt{q_\varphi} dW_\varphi(t).} \quad (21)$$

In this equation, $\lambda > 0$ is the relaxation rate of the phase process, $q_\varphi > 0$ is the phase-diffusion intensity, and $W_\varphi(t)$ is a standard Wiener process describing the random excitation of the phase. The process $W_\varphi(t)$ is assumed independent of the measurement noise $V(t)$. Together, (19) and (21) define the scalar continuous-time estimation problem treated by Kalman filtering in Chapter 3 [27, 28, 29].

8. Quadrature formulation

The previous sections described homodyne detection in terms of a phase difference between the signal and the local oscillator. For state-space modeling and quantum filtering, it is more useful to express the same measurement in terms of field quadratures [3, 4, 7, 25].

Let $\hat{a}_{\text{out}}(t)$ denote the annihilation operator of the output optical field measured by the homodyne detector. The subscript out indicates that the field is the output field after interaction with the optical system, for example after leaving an optical cavity. Consistently with the convention adopted in Chapter 4, the associated quadrature operators are defined *without* the $1/\sqrt{2}$ normalization:

$$\hat{q}_{\text{out}}(t) := \hat{a}_{\text{out}}(t) + \hat{a}_{\text{out}}^\dagger(t), \quad \hat{p}_{\text{out}}(t) := \frac{\hat{a}_{\text{out}}(t) - \hat{a}_{\text{out}}^\dagger(t)}{i}. \quad (22)$$

The quadrature $\hat{q}_{\text{out}}(t)$ is the amplitude-like quadrature, while $\hat{p}_{\text{out}}(t)$ is the phase-like quadrature. With this normalization, $[\hat{q}_{\text{out}}, \hat{p}_{\text{out}}] = 2i$ and the vacuum variance is $\langle \hat{q}_{\text{out}}^2 \rangle_{\text{vac}} = 1$.

Remark 1 (Convention used throughout the report) *We use the same quadrature normalization in this chapter, in Chapter 4, and in the appendix. The alternative symmetric convention with a factor $1/\sqrt{2}$ (used e.g. in [3, 5]) yields $[\hat{q}, \hat{p}] = i$ and vacuum variance $1/2$; the two differ by a global $\sqrt{2}$ rescaling that can be absorbed into the homodyne gain κ_m and the process-noise matrix G .*

Inverting (22),

$$\hat{a}_{\text{out}}(t) = \frac{\hat{q}_{\text{out}}(t) + i\hat{p}_{\text{out}}(t)}{2}, \quad \hat{a}_{\text{out}}^\dagger(t) = \frac{\hat{q}_{\text{out}}(t) - i\hat{p}_{\text{out}}(t)}{2}. \quad (23)$$

From (9), the operator selected by the local oscillator is proportional to

$$e^{-i\theta}\hat{a}_{\text{out}}(t) + e^{i\theta}\hat{a}_{\text{out}}^\dagger(t). \quad (24)$$

This motivates the definition of the rotated quadrature

$$\hat{X}_\theta(t) := e^{-i\theta}\hat{a}_{\text{out}}(t) + e^{i\theta}\hat{a}_{\text{out}}^\dagger(t). \quad (25)$$

The detailed algebraic derivation of the balanced homodyne difference current is standard in quantum optics and follows from the beam-splitter input–output relations [3, 4, 5]; see Appendix 51..

Substituting (23) into the definition (25) gives

$$\begin{aligned} \hat{X}_\theta(t) &= \frac{e^{-i\theta}}{2}(\hat{q}_{\text{out}} + i\hat{p}_{\text{out}}) + \frac{e^{i\theta}}{2}(\hat{q}_{\text{out}} - i\hat{p}_{\text{out}}) \\ &= \frac{1}{2}(e^{-i\theta} + e^{i\theta})\hat{q}_{\text{out}} + \frac{i}{2}(e^{-i\theta} - e^{i\theta})\hat{p}_{\text{out}} \\ &= \cos\theta \hat{q}_{\text{out}}(t) + \sin\theta \hat{p}_{\text{out}}(t), \end{aligned} \quad (26)$$

where we used $e^{-i\theta} + e^{i\theta} = 2\cos\theta$ and $e^{-i\theta} - e^{i\theta} = -2i\sin\theta$. Thus, a homodyne detector with local-oscillator phase θ measures the rotated quadrature

$$\boxed{\hat{X}_\theta(t) = \hat{q}_{\text{out}}(t) \cos\theta + \hat{p}_{\text{out}}(t) \sin\theta.} \quad (27)$$

For $\theta = 0$, the measured quadrature is $\hat{q}_{\text{out}}(t)$; for $\theta = \pi/2$, it is $\hat{p}_{\text{out}}(t)$; intermediate values correspond to rotated quadratures.

After detector calibration and after reducing the measured quantum quadrature to a classical measurement record, the homodyne signal is written as

$$dY_\theta(t) = \sqrt{\kappa_m} [q_{\text{out}}(t) \cos\theta + p_{\text{out}}(t) \sin\theta] dt + \sigma_v dV(t). \quad (28)$$

Here, $Y_\theta(t)$ is the integrated homodyne record obtained with homodyne angle θ ; $q_{\text{out}}(t)$ and $p_{\text{out}}(t)$ are the classical quadrature variables associated with the measured output field; $\kappa_m > 0$ is an effective measurement strength absorbing optical coupling, detector efficiency, and (residual) normalization constants; $\sigma_v > 0$ is the measurement-noise amplitude; and $V(t)$ is a standard Wiener process representing classical shot noise.

9. State-space measurement model of the homodyne detection

The quadrature formulation leads directly to the state-space measurement model used in the rest of the report. Suppose first that the internal optical mode is described by the real state vector

$$x(t) = \begin{bmatrix} q(t) \\ p(t) \end{bmatrix}. \quad (29)$$

The variables $q(t)$ and $p(t)$ are the internal quadratures of the optical system. For an optical cavity, they represent the amplitude-like and phase-like quadratures of the intracavity mode. They should not be confused with $q_{\text{out}}(t)$ and $p_{\text{out}}(t)$, which refer to the measured output field. A homodyne detector does not measure $q(t)$ and $p(t)$ separately. It measures a projection of the state along the direction fixed by the local-oscillator phase θ . In a normalized linear state-space model, the measurement equation is written as

$$dY(t) = C_\theta x(t) dt + \sigma_v dV(t), \quad (30)$$

where $Y(t)$ is the integrated measurement record, $V(t)$ is a standard Wiener process, and $\sigma_v > 0$ is the measurement-noise amplitude. The row matrix C_θ is the homodyne measurement matrix:

$$C_\theta = \sqrt{\kappa_m} \begin{bmatrix} \cos \theta & \sin \theta \end{bmatrix}. \quad (31)$$

The scalar product $C_\theta x(t)$ is explicitly

$$C_\theta x(t) = \sqrt{\kappa_m} (q(t) \cos \theta + p(t) \sin \theta). \quad (32)$$

This is the state-space version of the homodyne quadrature measurement. If the unknown optical phase is included as an additional state, the augmented state is

$$X_a(t) = \begin{bmatrix} q(t) \\ p(t) \\ \varphi(t) \end{bmatrix}. \quad (33)$$

The subscript a stands for augmented. The first two components are optical quadratures, and the third component is the hidden phase to be estimated. The corresponding homodyne measurement equation becomes

$$dY(t) = C_a X_a(t) dt + \sigma_v dV(t), \quad (34)$$

where C_a is the augmented homodyne measurement matrix. With the state ordering $X_a(t) = [q(t), p(t), \varphi(t)]^\top$, this matrix is

$$C_a = \sqrt{\kappa_m} \begin{bmatrix} \cos \theta & \sin \theta & 0 \end{bmatrix}. \quad (35)$$

The zero in the third column is structurally important. It means that the phase $\varphi(t)$ is not directly measured by the homodyne detector. The detector measures only a projection of the optical quadratures. Therefore, $\varphi(t)$ can be estimated only if it affects the dynamics of $q(t)$ and $p(t)$. This observation is the link between homodyne detection and observer design. In the scalar model, the phase is measured after local linearization of the homodyne current. In the cavity model, the phase first drives the intracavity quadratures, and only a quadrature projection is measured. The estimation of $\varphi(t)$ is therefore an observer problem: the hidden phase must be reconstructed from its indirect dynamical imprint on a noisy measurement record [30, 31, 27, 28].

Chapitre 3 – Classical Observer for Phase Estimation via Homodyne Detection

10. Introductory scalar phase-estimation model

The purpose of this chapter is to introduce the simplest classical estimation setup used in this report before moving to the full optical-cavity formulation. At this introductory level, the hidden phase $\varphi(t)$ of Chapter 2 is modeled directly as the scalar state to be estimated, and the internal dynamics of an optical cavity are not yet included. The single aim is to compare a deterministic Luenberger-type observer with a Kalman–Bucy filter on the simplest homodyne phase-estimation problem. The block G_φ represents the stochastic phase dynamics; the homodyne detector produces a noisy normalized classical measurement record $y_c(t)$, from which a classical observer reconstructs an estimate $\hat{\varphi}(t)$. This simplified model is useful because it isolates the estimation mechanism from the optical physics. In the cavity model, the phase will no longer be measured directly: it will first perturb the cavity quadratures, and only a projection of the output field will be observed. The present chapter should therefore be read as a preliminary calibration of the estimation tools on a deliberately oversimplified setup.

11. Continuous-time stochastic model

The phase is modeled as an Ornstein–Uhlenbeck process:

$$d\varphi(t) = -\lambda\varphi(t) dt + \sqrt{q_\varphi} dW_\varphi(t). \quad (36)$$

In this equation, $\lambda > 0$ is the phase relaxation rate, $q_\varphi > 0$ is the phase-diffusion intensity, and $W_\varphi(t)$ is a standard Wiener process. The drift term $-\lambda\varphi(t)$ pulls the phase back toward zero, while the stochastic term $\sqrt{q_\varphi} dW_\varphi(t)$ continuously re-excites the phase. This is the standard structure of a mean-reverting Gaussian Markov process [29, 28]. In engineering notation, one may formally write

$$\dot{\varphi}(t) = -\lambda\varphi(t) + \sqrt{q_\varphi} v(t),$$

where $v(t) = dW_\varphi(t)/dt$ is a formal white-noise process. This notation is convenient for interpretation, but the precise mathematical model is the stochastic differential equation (36). After homodyne linearization and normalization, the measurement record is modeled as

$$dy_c(t) = \varphi(t) dt + \sqrt{R_h} dV(t). \quad (37)$$

Here, $y_c(t)$ is the normalized integrated homodyne record, $V(t)$ is a standard Wiener process representing measurement shot noise, and $R_h > 0$ is the measurement-noise intensity. The processes $W_\varphi(t)$ and $V(t)$ are assumed independent. In ideal normalized homodyne units, one often writes

$$R_h = \frac{1}{4|\alpha|^2}, \quad (38)$$

where $|\alpha| > 0$ denotes the calibrated coherent signal amplitude, identified with the real signal amplitude A_s introduced in Chapter 2 after absorption of the detection efficiency and local-oscillator normalization constants. In the ideal limit $\eta = 1$ and $\sigma_h = 1$, the relation (38) follows

directly from $R_h = \sigma_h^2/K_h^2$ with $K_h = 2A_s$. A larger coherent amplitude therefore reduces the effective measurement noise, consistently with the standard interpretation of balanced homodyne detection [3, 4, 5]. Equivalently, using the formal white-noise notation $w(t) = dV(t)/dt$, the measurement is often written as

$$y_c(t) = \varphi(t) + \frac{1}{2|\alpha|}w(t).$$

This expression is useful for simulations and intuition, but (37) is the stochastic differential form used for filtering.

12. Deterministic observer and stochastic filter

The problem is now to reconstruct $\varphi(t)$ from the noisy measurement record $y_c(t)$. A natural first approach is to use a deterministic observer. For a linear state-space system

$$\dot{x}(t) = Ax(t) + Bu(t), \quad y(t) = Cx(t),$$

a Luenberger observer has the form

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L(y(t) - C\hat{x}(t)). \quad (39)$$

The vector $\hat{x}(t)$ is the estimated state, $u(t)$ is a known input, $y(t)$ is the measured output, and L is the observer gain. The quantity $y(t) - C\hat{x}(t)$ is the output prediction error, or innovation. If the pair (A, C) is observable, the gain L can be chosen so that the observer error dynamics are stable [31, 30]. For the scalar phase model, the deterministic observer becomes

$$\dot{\hat{\varphi}}(t) = -\lambda\hat{\varphi}(t) + L(y_c(t) - \hat{\varphi}(t)). \quad (40)$$

In this equation, $\hat{\varphi}(t)$ is the estimate of the true phase $\varphi(t)$, and $L > 0$ is a scalar observer gain. If the measurement were noise-free, the estimation error

$$e(t) = \varphi(t) - \hat{\varphi}(t)$$

would satisfy

$$\dot{e}(t) = -(\lambda + L)e(t).$$

Thus, increasing L makes the deterministic error converge faster. However, in the homodyne problem the measurement is noisy. A very large gain makes the observer react quickly, but it also amplifies the measurement noise. This is the main limitation of a deterministic observer in this setting. The Kalman filter addresses this limitation by treating both the phase dynamics and the measurement equation as stochastic. In general, for

$$dx(t) = Ax(t)dt + Bu(t)dt + GdW(t), \quad dy(t) = Cx(t)dt + DdV(t), \quad (41)$$

the matrices G and D describe how the process noise $W(t)$ and the measurement noise $V(t)$ enter the model. The Kalman–Bucy filter has the observer structure

$$d\hat{x}(t) = A\hat{x}(t)dt + Bu(t)dt + K(t)(dy(t) - C\hat{x}(t)dt), \quad (42)$$

where $K(t)$ is the Kalman gain. Unlike the Luenberger gain, $K(t)$ is not chosen by pole placement. It is computed from the error covariance

$$P(t) = \mathbb{E}\left[\left(x(t) - \hat{x}(t)\right)\left(x(t) - \hat{x}(t)\right)^\top\right],$$

which measures the uncertainty of the estimate. The covariance satisfies the Riccati equation

$$\dot{P}(t) = AP(t) + P(t)A^\top + GG^\top - P(t)C^\top (DD^\top)^{-1} CP(t), \quad (43)$$

and the corresponding gain is

$$K(t) = P(t)C^\top (DD^\top)^{-1}. \quad (44)$$

This is the classical continuous-time Kalman filtering construction [27, 28, 29]. Under linear-Gaussian assumptions, it gives the minimum-variance estimator.

13. Scalar Kalman–Bucy filter for the homodyne phase model

For the scalar phase-estimation problem, the state is

$$x(t) = \varphi(t).$$

Comparing (36) and (37) with the general system (41), one obtains

$$A = -\lambda, \quad C = 1, \quad G = \sqrt{q_\varphi}, \quad D = \sqrt{R_h}.$$

The Kalman–Bucy estimator is therefore

$$d\hat{\varphi}(t) = -\lambda\hat{\varphi}(t) dt + K(t)(dy_c(t) - \hat{\varphi}(t) dt). \quad (45)$$

The term

$$dy_c(t) - \hat{\varphi}(t) dt$$

is the innovation increment. It is the new measurement information after subtracting the predicted measurement $\hat{\varphi}(t) dt$. The scalar covariance is

$$P(t) = \mathbb{E} \left[(\varphi(t) - \hat{\varphi}(t))^2 \right].$$

It satisfies

$$\dot{P}(t) = -2\lambda P(t) + q_\varphi - \frac{P(t)^2}{R_h}. \quad (46)$$

Using $R_h = 1/(4|\alpha|^2)$, this becomes

$$\dot{P}(t) = -2\lambda P(t) + q_\varphi - 4|\alpha|^2 P(t)^2. \quad (47)$$

The Kalman gain is

$$K(t) = \frac{P(t)}{R_h} = 4|\alpha|^2 P(t). \quad (48)$$

This formula makes the role of the optical amplitude explicit. When the homodyne measurement is more informative, R_h is smaller and the filter can use a larger correction gain. When the measurement is noisy, the gain decreases and the estimator relies more strongly on the model prediction.

At steady state, if the covariance converges to a constant value P_∞ , it satisfies

$$0 = -2\lambda P_\infty + q_\varphi - \frac{P_\infty^2}{R_h}. \quad (49)$$

The physically meaningful solution is the positive root

$$P_\infty = R_h \left(-\lambda + \sqrt{\lambda^2 + \frac{q_\varphi}{R_h}} \right). \quad (50)$$

This expression gives the asymptotic phase-estimation uncertainty for the scalar homodyne model.

14. Discrete numerical implementation used in simulation

For numerical simulations, time is discretized as

$$t_k = k\Delta t,$$

where $\Delta t > 0$ is the time step and $k \in \{0, \dots, N\}$ is the discrete time index. The discrete estimate and covariance are denoted by

$$\hat{\varphi}_k \simeq \hat{\varphi}(t_k), \quad P_k \simeq P(t_k).$$

An Euler discretization of the Riccati equation gives

$$P_{k+1} = P_k + \Delta t \left(-2\lambda P_k + q_\varphi - \frac{P_k^2}{R_h} \right). \quad (51)$$

The discrete Kalman gain is then

$$K_{k+1} = \frac{P_{k+1}}{R_h}. \quad (52)$$

Remark 2 (Numerical stability of the explicit Riccati update) *The forward-Euler update (51) is conditionally stable. Linearizing the right-hand side of the Riccati equation (47) around a positive operating value P yields a local contraction rate*

$$\mu(P) = 2\lambda + \frac{2P}{R_h} = 2\lambda + 8|\alpha|^2 P,$$

so that the explicit Euler scheme preserves positivity and stability whenever

$$\Delta t \ll \frac{1}{2\lambda + 8|\alpha|^2 P_\infty}. \quad (53)$$

For larger time steps, P_{k+1} may become negative, which is non-physical and signals that the discretization should either be refined or replaced by a semi-implicit scheme of the form

$$P_{k+1} = P_k + \Delta t \left(-2\lambda P_{k+1} + q_\varphi - P_k P_{k+1} / R_h \right),$$

which is unconditionally non-negative under mild assumptions [29]. In all simulations reported below, condition (53) is satisfied with a safety margin of at least one order of magnitude.

If $\Delta y_k = y_c(t_{k+1}) - y_c(t_k)$ denotes the homodyne measurement increment, the Euler form of the Kalman–Bucy estimator is

$$\hat{\varphi}_{k+1} = \hat{\varphi}_k - \lambda \hat{\varphi}_k \Delta t + K_{k+1} (\Delta y_k - \hat{\varphi}_k \Delta t). \quad (54)$$

Equivalently, if the simulation stores a sampled normalized current

$$y_c[k] \simeq \frac{\Delta y_k}{\Delta t},$$

then (54) can be written as

$$\hat{\varphi}_{k+1} = \hat{\varphi}_k + \Delta t \left[-\lambda \hat{\varphi}_k + K_{k+1} (y_c[k] - \hat{\varphi}_k) \right]. \quad (55)$$

This is the recursion used for the classical homodyne-based phase estimator.

For comparison, the Luenberger-type update used in the same scalar setting is

$$\hat{\varphi}_{k+1}^L = \hat{\varphi}_k^L + \Delta t \left[-\lambda \hat{\varphi}_k^L + L (y_c[k] - \hat{\varphi}_k^L) \right], \quad (56)$$

where $L > 0$ is fixed. The two estimators therefore have the same correction structure. The difference is that the Kalman gain is computed from the noise statistics through the covariance P_k , whereas the Luenberger gain is imposed by design.

15. Numerical illustration and interpretation

Figure 2 illustrates a typical simulation of the scalar phase estimation problem. The true signal is the Ornstein–Uhlenbeck phase $\varphi(t)$. The homodyne measurement is a noisy normalized observation of this phase. The Luenberger observer uses a fixed correction gain L , while the Kalman filter uses the gain $K(t)$ obtained from the Riccati equation.

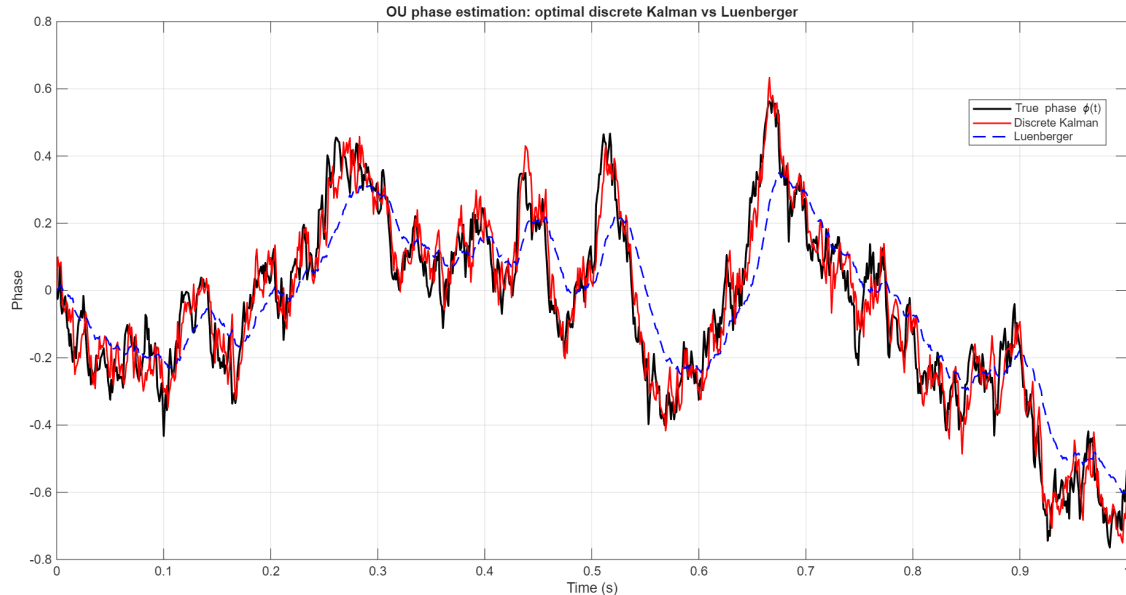


Figure 2: Comparison between a deterministic Luenberger observer and the scalar Kalman–Bucy estimator for homodyne-based phase estimation. The Kalman filter uses the noise intensities q_φ and R_h , whereas the Luenberger observer uses a fixed correction gain. *Simulation parameters:* $\lambda = 2.0\text{s}^{-1}$, $q_\varphi = 0.5\text{rad}^2/\text{s}$, $R_h = 2.5 \times 10^{-3}\text{s}$ (corresponding to $|\alpha| = 10$), fixed Luenberger gain $L = 10\text{s}^{-1}$, time step $\Delta t = 10^{-3}\text{s}$, total simulation horizon $T = 1\text{s}$, single realization. The CFL-type condition (53) is verified with $\Delta t (2\lambda + 8|\alpha|^2 P_\infty) \ll 1$.

Chapitre 4 – Optical cavity formulation as a phase-estimation problem

The previous chapters introduced two ingredients that will now be combined. First, balanced homodyne detection converts an optical quadrature into a classical stochastic measurement record. Second, the unknown optical phase was modeled as a stochastic process and estimated from noisy observations. The purpose of the present chapter is to embed this phase-estimation problem inside the dynamics of an open optical cavity. The important change is the following. In a coherent mode-tracking problem, one may compare the phase of an intracavity field with the phase of another field, for instance through quantities such as $\arg(a(t))$ and $\arg(\tilde{a}(t))$. This is a useful diagnostic of cavity-mode tracking, but it is not yet the estimation of an independent hidden phase. Here, instead, the unknown quantity is an external stochastic phase

$$\varphi(t),$$

which enters the coherent drive of the cavity. The cavity transforms this hidden phase into fluctuations of its internal quadratures, and the homodyne detector measures only a noisy projection of those quadratures. The estimation problem is therefore indirect: the phase is not measured directly, but must be reconstructed from its dynamical imprint on the cavity output. This chapter derives the corresponding state-space model. We first introduce the open-cavity Hamiltonian, eliminate the external bath under the Markov approximation, include a phase-modulated coherent drive, and linearize the model around a steady operating point. We then rewrite the dynamics in quadrature form, augment the state with an Ornstein–Uhlenbeck phase process, and connect the resulting system to the homodyne measurement model. The construction follows the standard input-output description of open quantum optical systems [4, 3, 5] and leads to a continuous-time linear-Gaussian model suitable for observability analysis and Kalman–Bucy filtering [27, 28, 29, 31].

16. Open single-mode cavity

We consider a single optical cavity mode coupled to an external electromagnetic continuum, following the standard input–output formalism of open quantum optical systems [32, 4, 3, 33]. The intracavity annihilation operator is denoted by $\hat{a}(t)$, with $[\hat{a}, \hat{a}^\dagger] = 1$. The bath is a continuum of bosonic modes $\hat{b}(\nu)$ indexed by the bath frequency ν (the symbol ν avoids confusion with the cavity–laser detuning introduced later), with $[\hat{b}(\nu), \hat{b}^\dagger(\nu')] = \delta(\nu - \nu')$. The total Hamiltonian decomposes as $\hat{H} = \hat{H}_S + \hat{H}_B + \hat{H}_{\text{int}} + \hat{H}_{\text{drv}}$, where

$$\hat{H}_S = \hbar\omega_c \hat{a}^\dagger \hat{a}, \quad \hat{H}_B = \int_{\mathbb{R}} \hbar\nu \hat{b}^\dagger(\nu) \hat{b}(\nu) d\nu, \quad (57)$$

$$\hat{H}_{\text{int}} = i\hbar \int_{\mathbb{R}} \left(\kappa(\nu) \hat{a}^\dagger \hat{b}(\nu) - \kappa^*(\nu) \hat{b}^\dagger(\nu) \hat{a} \right) d\nu, \quad (58)$$

in the rotating-wave approximation [33, 4]. Here ω_c is the bare cavity resonance, $\kappa(\nu)$ is the frequency-dependent coupling describing photon exchange through the partially transmitting mirror, and $\hat{H}_{\text{drv}}(t)$ is the coherent drive

$$\hat{H}_{\text{drv}}(t) = i\hbar \left(\mathcal{E}(t) \hat{a}^\dagger - \mathcal{E}^*(t) \hat{a} \right), \quad \mathcal{E}(t) = \epsilon e^{-i\omega_L t} e^{i\varphi(t)}, \quad (59)$$

with nominal drive amplitude $\epsilon > 0$, laser frequency ω_L , and stochastic optical phase $\varphi(t)$ to be estimated. Eliminating the bath under the flat-coupling (Markov) approximation around the cavity resonance yields the standard quantum Langevin equation [32, 4, 3]:

$$\boxed{\frac{d\hat{a}(t)}{dt} = -\left(i\omega_c + \frac{\gamma}{2}\right)\hat{a}(t) + \mathcal{E}(t) + \sqrt{\gamma}\hat{a}_{\text{in}}(t).} \quad (60)$$

The cavity linewidth $\gamma > 0$ is the energy decay rate, and $\hat{a}_{\text{in}}(t)$ is the quantum input field, with zero mean in vacuum and $[\hat{a}_{\text{in}}(t), \hat{a}_{\text{in}}^\dagger(t')] = \delta(t - t')$. The detailed derivation (Heisenberg equation, formal solution of the bath modes, substitution back) is standard and can be found in [4, Ch. 3] and [3, Ch. 5].

17. Rotating frame and phase-modulated drive

It is convenient to move to the frame rotating at the laser frequency ω_L by setting $\hat{a}_r(t) = e^{i\omega_L t}\hat{a}(t)$; to avoid heavy notation, we again denote this operator by $\hat{a}(t)$ [3, Ch. 5]. Writing the cavity–laser detuning $\omega := \omega_c - \omega_L$ (so that $\omega = 0$ is resonant driving), the Langevin equation (60) becomes

$$\frac{d\hat{a}(t)}{dt} = \left(-\frac{\gamma}{2} - i\omega\right)\hat{a}(t) + \epsilon e^{i\varphi(t)} + \sqrt{\gamma}\hat{a}_{\text{in}}(t). \quad (61)$$

Mean-field linearization. Equation (61) is nonlinear in φ through the factor $e^{i\varphi(t)}$. Following the standard mean-field expansion of cavity QED [3, 4, 33], we linearize around the operating point $\varphi = 0$ using $e^{i\varphi(t)} \simeq 1 + i\varphi(t)$ (valid for $|\varphi| \ll 1$; see assumption (A1) of Chapter 5) and decompose the field as $\hat{a}(t) = \alpha + \delta\hat{a}(t)$, where α is the steady-state amplitude solving $\left(-\frac{\gamma}{2} - i\omega\right)\alpha + \epsilon = 0$:

$$\boxed{\alpha = \frac{\epsilon}{\frac{\gamma}{2} + i\omega}.} \quad (62)$$

Subtracting the steady-state equation from (61) gives the linearized fluctuation dynamics

$$\frac{d\delta\hat{a}(t)}{dt} = \left(-\frac{\gamma}{2} - i\omega\right)\delta\hat{a}(t) + i\epsilon\varphi(t) + \sqrt{\gamma}\hat{a}_{\text{in}}(t). \quad (63)$$

From now on, $\hat{a}(t)$ denotes the fluctuation $\delta\hat{a}(t)$ unless stated otherwise; the state variables of the next section describe fluctuations around the coherent operating point, not the full optical amplitude.

18. Quadrature state-space model

We now rewrite the complex fluctuation model in real quadrature coordinates. Consistently with the convention used in the rest of the report, we define

$$\hat{q}(t) = \hat{a}(t) + \hat{a}^\dagger(t), \quad \hat{p}(t) = \frac{\hat{a}(t) - \hat{a}^\dagger(t)}{i}. \quad (64)$$

Equivalently,

$$\hat{a}(t) = \frac{\hat{q}(t) + i\hat{p}(t)}{2}, \quad \hat{a}^\dagger(t) = \frac{\hat{q}(t) - i\hat{p}(t)}{2}. \quad (65)$$

With this normalization, $[\hat{q}, \hat{p}] = 2i$. Other quantum-optics conventions include a factor $1/\sqrt{2}$; this only changes numerical normalization constants and is absorbed here into the effective gains.

The input quadratures are defined in the same convention:

$$\hat{q}_{\text{in}}(t) = \hat{a}_{\text{in}}(t) + \hat{a}_{\text{in}}^\dagger(t), \quad \hat{p}_{\text{in}}(t) = \frac{\hat{a}_{\text{in}}(t) - \hat{a}_{\text{in}}^\dagger(t)}{i}. \quad (66)$$

Separating real and imaginary parts in (63) gives the linear quadrature model

$$\frac{d\hat{q}(t)}{dt} = -\frac{\gamma}{2}\hat{q}(t) + \omega\hat{p}(t) + \sqrt{\gamma}\hat{q}_{\text{in}}(t), \quad (67)$$

$$\frac{d\hat{p}(t)}{dt} = -\omega\hat{q}(t) - \frac{\gamma}{2}\hat{p}(t) + g_\varphi\varphi(t) + \sqrt{\gamma}\hat{p}_{\text{in}}(t). \quad (68)$$

The coefficient g_φ is the effective phase-to-quadrature coupling. For the drive convention above, it is proportional to the drive amplitude ϵ , and equivalently to the operating amplitude α through (62). The matrix $B_\varphi = [0, g_\varphi]^\top$ is the phase-input matrix; it encodes the fact that the hidden phase drives the p -quadrature only (resonant convention), via the phase-to-quadrature coupling g_φ . Convention-dependent factors are absorbed into g_φ . The phase convention has been chosen so that the linearized phase perturbation enters the p -quadrature. For estimation and observability analysis, we pass from the operator model to the corresponding real linear stochastic model for the quadrature coordinates. The reduced cavity state is

$$x(t) = \begin{bmatrix} q(t) \\ p(t) \end{bmatrix}. \quad (69)$$

The control-oriented fluctuation dynamics are written as

$$dx(t) = Ax(t) dt + B_\varphi\varphi(t) dt + G dW_x(t), \quad (70)$$

where

$$A = \begin{bmatrix} -\gamma/2 & \omega \\ -\omega & -\gamma/2 \end{bmatrix}, \quad B_\varphi = \begin{bmatrix} 0 \\ g_\varphi \end{bmatrix}. \quad (71)$$

The process $W_x(t)$ is a two-dimensional Wiener process representing the effective vacuum-driven quadrature noise after the linear Gaussian reduction. For a normalized vacuum-limited model, one may take $G = \sqrt{\gamma}I_2$. More generally, G is kept as the process-noise input matrix. The information pathway is now explicit:

$$\varphi(t) \longrightarrow p(t) \longrightarrow (q(t), p(t)) \longrightarrow \text{homodyne record}.$$

The phase does not appear as a measured output. It first drives the internal cavity dynamics.

19. Stochastic phase model

The hidden phase is modeled by an Ornstein–Uhlenbeck process,

$$d\varphi(t) = -\lambda\varphi(t) dt + \sqrt{q_\varphi} dW_\varphi(t), \quad (72)$$

where $\lambda > 0$ is the relaxation rate and $q_\varphi > 0$ is the phase diffusion intensity. The Wiener process $W_\varphi(t)$ is assumed independent of the measurement noise. This model is appropriate when the phase has finite correlation time: the deterministic term pulls the phase back to zero, while the stochastic term continuously re-excites it. The stationary variance of the OU phase is

$$\text{Var}[\varphi(t)] = \frac{q_\varphi}{2\lambda}, \quad (73)$$

when the process is initialized in stationarity. Thus, increasing q_φ makes the phase more volatile, whereas increasing λ shortens its memory and reduces its stationary variance.

20. Single-cavity augmented model

We now augment the cavity quadratures with the hidden phase. The augmented state is

$$X_a(t) = \begin{bmatrix} q(t) \\ p(t) \\ \varphi(t) \end{bmatrix}. \quad (74)$$

The subscript a means “augmented”. The first two components are the cavity quadratures, and the third component is the stochastic phase to be estimated. Combining (70) and (72) gives

$$dX_a(t) = A_a X_a(t) dt + G_a dW_a(t), \quad (75)$$

with

$$A_a = \begin{bmatrix} -\gamma/2 & \omega & 0 \\ -\omega & -\gamma/2 & g_\varphi \\ 0 & 0 & -\lambda \end{bmatrix}. \quad (76)$$

The augmented Wiener process is

$$W_a(t) = \begin{bmatrix} W_x(t) \\ W_\varphi(t) \end{bmatrix}. \quad (77)$$

If the cavity quadrature noise is modeled as vacuum-limited and independent of the phase noise, then

$$G_a = \begin{bmatrix} \sqrt{\gamma} & 0 & 0 \\ 0 & \sqrt{\gamma} & 0 \\ 0 & 0 & \sqrt{q_\varphi} \end{bmatrix}. \quad (78)$$

In a more general reduced model, the upper-left 2×2 block can be replaced by the appropriate process-noise matrix G .

21. Homodyne measurement model

The input-output relation for a single-sided cavity is

$$\hat{a}_{\text{out}}(t) = \hat{a}_{\text{in}}(t) + \sqrt{\gamma} \hat{a}(t). \quad (79)$$

A balanced homodyne detector measures a rotated quadrature of this output field. If θ is the local-oscillator phase, the measured output quadrature is proportional to

$$\hat{q}_{\text{out}}(t) \cos \theta + \hat{p}_{\text{out}}(t) \sin \theta. \quad (80)$$

After calibration by the known detector gain and after grouping the vacuum shot-noise contribution into an effective classical measurement noise, the measurement record is modeled as

$$dY(t) = C_a X_a(t) dt + \sigma_v dV(t), \quad (81)$$

where $V(t)$ is a standard Wiener process and $\sigma_v > 0$ is the measurement-noise amplitude. The measurement matrix is

$$C_a = \sqrt{\kappa_m} \begin{bmatrix} \cos \theta & \sin \theta & 0 \end{bmatrix}. \quad (82)$$

The parameter $\kappa_m > 0$ is an effective homodyne measurement strength. It absorbs optical coupling, detector efficiency, local-oscillator normalization, and calibration constants. In an ideal normalized model, κ_m may be chosen proportional to the physical decay rate γ , but it is useful to keep the

two symbols distinct: γ controls the cavity dynamics, whereas κ_m controls the scale of the classical measurement channel. The zero in the third column of C_a is structurally essential:

$$C_a e_\varphi = 0, \quad e_\varphi = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (83)$$

Thus the homodyne detector does not measure $\varphi(t)$ directly. It measures only a quadrature projection of the cavity field. The phase can be estimated only because it affects the dynamics of $q(t)$ and $p(t)$. The matrices G_s and G_o are the process-noise input matrices for the sensing and auxiliary cavities. In the vacuum-limited normalized model, $G_s = \sqrt{\gamma_s} I_2$ and $G_o = \sqrt{\gamma_o} I_2$, reflecting that vacuum input fluctuations enter both quadratures with amplitude $\sqrt{\gamma}$. The augmented matrix $G_Q = \text{diag}(G_s, G_o, \sqrt{q_\varphi})$ collects the cavity quadrature noises and the OU phase diffusion in a block-diagonal structure. The resulting single-cavity phase-estimation model is therefore

$$\begin{cases} dX_a(t) = A_a X_a(t) dt + G_a dW_a(t), \\ dY(t) = C_a X_a(t) dt + \sigma_v dV(t). \end{cases} \quad (84)$$

This is the model used for linear observability analysis and Kalman–Bucy filtering.

22. Interpretation as an estimation problem

The variable to be reconstructed is not the phase of the cavity mode itself. The quantity of interest is the external latent phase process $\varphi(t)$ appearing in the coherent drive. The cavity acts as a dynamical transducer: it converts the hidden phase into quadrature fluctuations. The homodyne detector then observes only a noisy one-dimensional projection of those fluctuations.

The estimation problem can therefore be stated as follows:

Given the homodyne measurement record $Y(t)$, estimate the hidden stochastic phase $\varphi(t)$ from its indirect effect on the cavity quadratures.

This formulation is the bridge between quantum-optical modeling and classical state estimation. Once the reduced linear-Gaussian model has been obtained, the phase-estimation performance is governed by the joint structure of the drift matrix A_a , the process-noise matrix G_a , the homodyne matrix C_a , and the measurement-noise level σ_v^2 .

23. Augmented two-cavity preprocessing model

We now introduce an augmented architecture containing a sensing cavity and an auxiliary cavity. The purpose of this model is not to claim, by itself, a full hardware-realizable quantum network. Its purpose is to isolate a precise control-theoretic question: can an auxiliary dynamical subsystem reshape the pathway through which information about $\varphi(t)$ reaches the homodyne measurement? The sensing cavity is denoted by the subscript s , and the auxiliary cavity by the subscript o . Their quadrature states are

$$x_s(t) = \begin{bmatrix} q_s(t) \\ p_s(t) \end{bmatrix}, \quad x_o(t) = \begin{bmatrix} q_o(t) \\ p_o(t) \end{bmatrix}. \quad (85)$$

The hidden phase $\varphi(t)$ drives only the sensing cavity. The full augmented state is

$$X_Q(t) = \begin{bmatrix} q_s(t) \\ p_s(t) \\ q_o(t) \\ p_o(t) \\ \varphi(t) \end{bmatrix} = \begin{bmatrix} x_s(t) \\ x_o(t) \\ \varphi(t) \end{bmatrix}. \quad (86)$$

The uncoupled local dynamics are described by

$$A_s = \begin{bmatrix} -\gamma_s/2 & \omega_s \\ -\omega_s & -\gamma_s/2 \end{bmatrix}, \quad A_o = \begin{bmatrix} -\gamma_o/2 & \omega_o \\ -\omega_o & -\gamma_o/2 \end{bmatrix}. \quad (87)$$

Here, $\gamma_s, \gamma_o > 0$ are the linewidths of the sensing and auxiliary cavities, while ω_s and ω_o are their rotating-frame detunings. The hidden phase enters the sensing cavity through

$$B_s = \begin{bmatrix} 0 \\ g_\varphi \end{bmatrix}. \quad (88)$$

To keep the direction of the coupling unambiguous, we write $K_{s \leftarrow o}$ for the block multiplying x_o in the sensing-cavity equation, and $K_{o \leftarrow s}$ for the block multiplying x_s in the auxiliary-cavity equation. Thus,

$$dx_s(t) = (A_s x_s(t) + K_{s \leftarrow o} x_o(t) + B_s \varphi(t)) dt + G_s dW_s(t), \quad (89)$$

$$dx_o(t) = (A_o x_o(t) + K_{o \leftarrow s} x_s(t)) dt + G_o dW_o(t). \quad (90)$$

The matrices $K_{s \leftarrow o}$ and $K_{o \leftarrow s}$ describe the linear dynamical coupling between the two cavities. In numerical studies, a simple isotropic surrogate is often sufficient:

$$K_{s \leftarrow o} = k_{s \leftarrow o} I_2, \quad K_{o \leftarrow s} = k_{o \leftarrow s} I_2. \quad (91)$$

This choice should be understood as a reduced linear preprocessing model. If one wants to enforce a specific Hamiltonian beam-splitter interaction or full quantum physical realizability, the coupling blocks must be replaced by the corresponding physically realizable quadrature matrices. For linear quantum systems, physical realizability imposes algebraic constraints on the drift and noise matrices, for example preservation of the canonical commutation relations [23, 34, 24, 25]. In the present chapter, the augmented model is used primarily as a controlled state-space surrogate for studying information flow and estimation trade-offs [12, 13, 14, 15]. Combining (89), (90), and the OU phase equation gives

$$dX_Q(t) = A_Q X_Q(t) dt + G_Q dW_Q(t), \quad (92)$$

where

$$A_Q = \begin{bmatrix} A_s & K_{s \leftarrow o} & B_s \\ K_{o \leftarrow s} & A_o & 0 \\ 0 & 0 & -\lambda \end{bmatrix}. \quad (93)$$

In the isotropic case (91), this becomes

$$A_Q = \begin{bmatrix} -\gamma_s/2 & \omega_s & k_{s \leftarrow o} & 0 & 0 \\ -\omega_s & -\gamma_s/2 & 0 & k_{s \leftarrow o} & g_\varphi \\ k_{o \leftarrow s} & 0 & -\gamma_o/2 & \omega_o & 0 \\ 0 & k_{o \leftarrow s} & -\omega_o & -\gamma_o/2 & 0 \\ 0 & 0 & 0 & 0 & -\lambda \end{bmatrix}. \quad (94)$$

The augmented noise vector is

$$W_Q(t) = \begin{bmatrix} W_s(t) \\ W_o(t) \\ W_\varphi(t) \end{bmatrix}, \quad (95)$$

and the noise matrix is

$$G_Q = \text{diag}(G_s, G_o, \sqrt{q_\varphi}). \quad (96)$$

For vacuum-limited local cavity noises, one may take $G_s = \sqrt{\gamma_s} I_2$ and $G_o = \sqrt{\gamma_o} I_2$. More general process-noise models can be used without changing the structure of (92).

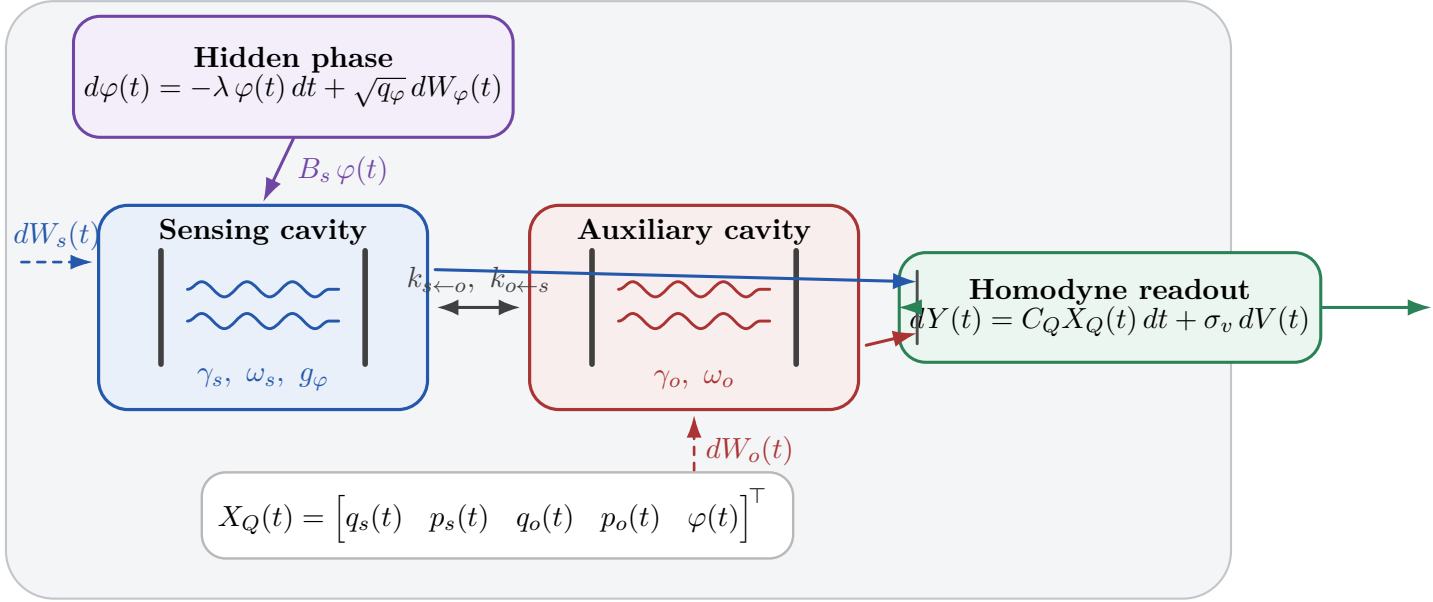


Figure 3: Augmented two-cavity architecture. The hidden Ornstein–Uhlenbeck phase $\varphi(t)$ drives the sensing cavity through $B_s \varphi(t)$; an auxiliary cavity is coupled to the sensing cavity via the Hamiltonian coupling matrices $k_{s \leftarrow o}$, $k_{o \leftarrow s}$; and the homodyne measurement combines the accessible quadratures of both cavities, with measurement matrix C_Q and shot-noise increment $\sigma_v dV(t)$. The full augmented state $X_Q(t)$ is shown below the diagram.

24. Two-cavity homodyne readout

The homodyne measurement is again a linear projection of accessible output quadratures. To make the optical meaning clear, we parametrize the readout as a weighted sum of a sensing-cavity quadrature and an auxiliary-cavity quadrature:

$$dY(t) = C_Q X_Q(t) dt + \sigma_v dV(t), \quad (97)$$

with

$$C_Q = \sqrt{\kappa_m} \begin{bmatrix} \ell_s \cos \theta_s & \ell_s \sin \theta_s & \ell_o \cos \theta_o & \ell_o \sin \theta_o & 0 \end{bmatrix}. \quad (98)$$

Here, θ_s and θ_o are the homodyne quadrature angles associated with the two output channels, and $\ell_s, \ell_o \in \mathbb{R}$ are readout weights. In a normalized lossless optical mixing model, one may impose $\ell_s^2 + \ell_o^2 = 1$. If only the sensing-cavity output is measured, then $\ell_s = 1$ and $\ell_o = 0$. If an optical network mixes the two outputs before detection, then both weights may be nonzero. Expanded explicitly,

$$C_Q X_Q(t) = \sqrt{\kappa_m} [\ell_s (q_s(t) \cos \theta_s + p_s(t) \sin \theta_s) + \ell_o (q_o(t) \cos \theta_o + p_o(t) \sin \theta_o)]. \quad (99)$$

Again, the last coefficient is zero:

$$C_Q e_\varphi = 0, \quad e_\varphi = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}. \quad (100)$$

The phase is not measured directly. It must pass through the sensing cavity, then possibly through the auxiliary dynamics, and finally through the homodyne projection. The complete two-cavity preprocessing model is

$$\boxed{\begin{aligned} dX_Q(t) &= A_Q X_Q(t) dt + G_Q dW_Q(t), \\ dY(t) &= C_Q X_Q(t) dt + \sigma_v dV(t). \end{aligned}} \quad (101)$$

25. Role of the auxiliary cavity

The auxiliary cavity should be interpreted as a dynamical preprocessing subsystem. It does not observe $\varphi(t)$ directly, and it does not contain an independent copy of the phase. Its only access to phase information comes through its coupling to the sensing cavity. The relevant pathway is

$$\varphi(t) \longrightarrow p_s(t) \longrightarrow x_s(t) \longleftrightarrow x_o(t) \longrightarrow Y(t).$$

This pathway can be beneficial or detrimental. It can improve estimation if the auxiliary dynamics rotate or amplify phase-dependent components into the measured quadrature direction. It can degrade estimation if the auxiliary mode injects additional noise, creates poorly observable modes, or transfers energy away from the measured phase-sensitive direction. Consequently, the augmented architecture should not be judged only by the presence of an additional cavity. Its value depends on the combined structure of A_Q , G_Q , C_Q , and the stochastic parameters $(\lambda, q_\varphi, \sigma_v^2)$. This is why the next chapter studies both directional observability and filtering performance. The central question is not whether the auxiliary cavity is always better, but in which dynamical regimes it increases useful phase visibility faster than it increases the effective noise burden.

Chapitre 5 – Cavity Phase Estimation: Observability Limits and Regime-Dependent Preprocessing Trade-Offs

26. Introduction

Optical and microwave cavities play a central role in modern sensing and continuous-measurement architectures, from cavity optomechanics to quantum-limited detection and quantum sensing [1, 5, 35, 3]. In such systems, the physical quantity of interest is often not measured directly. Instead, it is encoded into internal cavity variables and must be reconstructed from noisy continuous measurements. This naturally leads to a hidden-state estimation problem, for which Kalman–Bucy filtering and continuous-time stochastic estimation provide the classical reference framework [27, 28, 29, 18]. In the previous chapter, the hidden optical phase was embedded into a linearized cavity model. The phase is denoted throughout this chapter by

$$\varphi(t),$$

where $t \geq 0$ is time. It is a real-valued stochastic process entering the coherent drive of the cavity. The key point is that $\varphi(t)$ is not directly observed. It first perturbs the intracavity quadratures, and only then appears indirectly in the homodyne photocurrent. Therefore, before designing or simulating a filter, one should ask a structural question:

Under what conditions is the hidden phase direction effectively visible from the homodyne measurement channel?

To answer this question, we introduce a directional phase-observability metric. Let $X_a(t)$ denote the augmented single-cavity state,

$$X_a(t) = \begin{bmatrix} q(t) \\ p(t) \\ \varphi(t) \end{bmatrix},$$

where $q(t)$ and $p(t)$ are respectively the amplitude-like and phase-like intracavity quadratures. The subscript a stands for “augmented”, because the physical cavity quadratures have been augmented with the hidden phase state. Let A_a be the drift matrix of this augmented linear system and C_a the corresponding homodyne measurement row. Over a finite observation horizon $[0, T]$, the continuous-time observability Gramian is

$$W_o(T) = \int_0^T e^{A_a^\top t} C_a^\top C_a e^{A_a t} dt.$$

Here, the subscript o stands for observability. The canonical basis vector associated with the phase component is

$$e_\varphi = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

We define the directional phase-observability metric by

$$J_\varphi(T) = e_\varphi^\top W_o(T) e_\varphi. \quad (102)$$

This quantity measures the output energy generated by an initial perturbation along the phase direction. It is therefore a structural proxy for the visibility of the hidden phase through homodyne detection [30, 31]. The main message of this chapter is that $J_\varphi(T)$ is useful, but not universal. Within a fixed dynamical family, it explains clear trends in the phase-estimation root-mean-square error. Across mixed stochastic regimes, however, the Ornstein–Uhlenbeck relaxation rate, the phase diffusion intensity, the measurement noise, and the homodyne geometry introduce additional independent effects. Consequently, the estimation error cannot, in general, be reduced to a single scalar function of $J_\varphi(T)$. We also study an augmented architecture in which the sensing cavity is coupled to an auxiliary cavity before measurement. This auxiliary subsystem is used as a reduced linear model of dynamical preprocessing. It can improve the visibility of the hidden phase by rotating or redistributing phase-dependent components toward the measured quadrature. However, it can also introduce additional noise channels or poorly observable modes. Its effect is therefore regime-dependent. This perspective is consistent with the broader literature on linear quantum systems, coherent observers, and coherent-classical estimation [25, 12, 13, 14, 15].

27. Single-cavity phase-estimation model

We begin with the reduced single-cavity model derived in the previous chapter. The intracavity annihilation operator, after moving to the frame rotating at the laser frequency and after recentering around the coherent operating point, is denoted by $a(t)$. Its adjoint is $a^\dagger(t)$. The corresponding quadratures are

$$q(t) = a(t) + a^\dagger(t), \quad p(t) = \frac{a(t) - a^\dagger(t)}{i}. \quad (103)$$

The quadrature $q(t)$ is amplitude-like, while $p(t)$ is phase-like. The real cavity state is

$$x(t) = \begin{bmatrix} q(t) \\ p(t) \end{bmatrix}. \quad (104)$$

The cavity linewidth is denoted by $\gamma > 0$. It represents the energy decay rate of the cavity mode through its coupling mirror. The detuning between the cavity resonance and the laser frequency is denoted by ω . The hidden phase $\varphi(t)$ enters the linearized cavity dynamics through the effective phase-to-quadrature coupling g_φ . With the phase convention used here, this coupling acts on the p -quadrature. The reduced continuous-time model is

$$dx(t) = Ax(t) dt + B_\varphi \varphi(t) dt + G_x dW_x(t), \quad (105)$$

where

$$A = \begin{bmatrix} -\gamma/2 & \omega \\ -\omega & -\gamma/2 \end{bmatrix}, \quad B_\varphi = \begin{bmatrix} 0 \\ g_\varphi \end{bmatrix}. \quad (106)$$

The matrix G_x maps the effective quadrature process noise into the cavity state. The process $W_x(t)$ is a two-dimensional Wiener process representing the reduced vacuum-driven quadrature fluctuations. In the simplest vacuum-limited normalized model, one may take

$$G_x = \sqrt{\gamma} I_2,$$

where I_2 is the 2×2 identity matrix. The hidden phase is modeled as an Ornstein–Uhlenbeck process:

$$d\varphi(t) = -\lambda\varphi(t) dt + \sqrt{q_\varphi} dW_\varphi(t). \quad (107)$$

Here, $\lambda > 0$ is the phase relaxation rate, $q_\varphi > 0$ is the phase diffusion intensity, and $W_\varphi(t)$ is a scalar standard Wiener process. The larger q_φ is, the more strongly the phase is randomly excited. The larger λ is, the faster the phase relaxes back toward zero. When the process is initialized in stationarity, its variance is

$$\text{Var}[\varphi(t)] = \frac{q_\varphi}{2\lambda}.$$

Combining the cavity quadratures and the phase gives the augmented state

$$X_a(t) = \begin{bmatrix} q(t) \\ p(t) \\ \varphi(t) \end{bmatrix}. \quad (108)$$

The corresponding augmented stochastic model is

$$dX_a(t) = A_a X_a(t) dt + G_a dW_a(t), \quad (109)$$

with

$$A_a = \begin{bmatrix} -\gamma/2 & \omega & 0 \\ -\omega & -\gamma/2 & g_\varphi \\ 0 & 0 & -\lambda \end{bmatrix}. \quad (110)$$

The augmented noise process is

$$W_a(t) = \begin{bmatrix} W_x(t) \\ W_\varphi(t) \end{bmatrix}.$$

If $G_x = \sqrt{\gamma}I_2$, then

$$G_a = \begin{bmatrix} \sqrt{\gamma} & 0 & 0 \\ 0 & \sqrt{\gamma} & 0 \\ 0 & 0 & \sqrt{q_\varphi} \end{bmatrix}. \quad (111)$$

This model makes the information pathway explicit:

$$\varphi(t) \longrightarrow p(t) \longrightarrow (q(t), p(t)) \longrightarrow \text{homodyne record}.$$

The phase does not appear directly at the detector. It becomes visible only through the dynamics of the cavity quadratures.

28. Homodyne measurement model

The cavity output is monitored by balanced homodyne detection. The homodyne angle is denoted by θ . It is the phase of the local oscillator and selects which quadrature direction is measured. In the reduced state-space model, the homodyne measurement of the cavity state $x(t)$ is written as

$$dY(t) = C_\theta x(t) dt + \sigma_v dV(t), \quad (112)$$

where $Y(t)$ is the integrated classical homodyne record, $V(t)$ is a scalar standard Wiener process representing measurement shot noise, and $\sigma_v > 0$ is the measurement-noise amplitude. The row vector C_θ is

$$C_\theta = \sqrt{\kappa_m} \begin{bmatrix} \cos \theta & \sin \theta \end{bmatrix}. \quad (113)$$

The parameter $\kappa_m > 0$ is the effective homodyne measurement gain. It absorbs the optical coupling, detector efficiency, local-oscillator normalization, and calibration constants. We keep κ_m distinct from the physical linewidth γ : γ governs the cavity dynamics, whereas κ_m governs the scale of the measurement channel. In augmented coordinates, the measurement equation becomes

$$dY(t) = C_a X_a(t) dt + \sigma_v dV(t), \quad (114)$$

with

$$C_a = \sqrt{\kappa_m} \begin{bmatrix} \cos \theta & \sin \theta & 0 \end{bmatrix}. \quad (115)$$

The zero in the third column is central. It means that the detector does not measure $\varphi(t)$ directly. If

$$e_\varphi = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

then

$$C_a e_\varphi = 0. \quad (116)$$

Therefore, any information about the phase must arise from the dynamical coupling between $\varphi(t)$ and the cavity quadratures.

29. Kalman–Bucy estimation baseline

The reduced phase-estimation model is linear and Gaussian:

$$dX_a(t) = A_a X_a(t) dt + G_a dW_a(t), \quad dY(t) = C_a X_a(t) dt + \sigma_v dV(t). \quad (117)$$

The process covariance and measurement covariance are

$$Q = G_a G_a^\top, \quad R = \sigma_v^2. \quad (118)$$

The matrix Q quantifies the strength of the state noise, including both vacuum quadrature noise and OU phase diffusion. The scalar R is the variance intensity of the homodyne measurement noise.

The optimal continuous-time linear-Gaussian estimator is the Kalman–Bucy filter [27, 28, 29]. Its estimate is denoted by

$$\hat{X}_a(t) = \begin{bmatrix} \hat{q}(t) \\ \hat{p}(t) \\ \hat{\varphi}(t) \end{bmatrix}.$$

It evolves according to

$$d\hat{X}_a(t) = A_a \hat{X}_a(t) dt + K(t) (dY(t) - C_a \hat{X}_a(t) dt), \quad (119)$$

where

$$K(t) = P(t) C_a^\top R^{-1} \quad (120)$$

is the Kalman gain. The matrix $P(t)$ is the estimation-error covariance:

$$P(t) = \mathbb{E} \left[\left(X_a(t) - \hat{X}_a(t) \right) \left(X_a(t) - \hat{X}_a(t) \right)^\top \right].$$

It satisfies the Riccati equation

$$\dot{P}(t) = A_a P(t) + P(t) A_a^\top + Q - P(t) C_a^\top R^{-1} C_a P(t). \quad (121)$$

The phase-estimation error is the scalar difference

$$\varphi(t) - \hat{\varphi}(t).$$

In simulations, the phase root-mean-square error is computed after a burn-in time T_{burn} , in order to reduce initialization transients. If $T_b > 0$ is the averaging duration after burn-in, then

$$\text{RMSE}(\varphi) = \left(\frac{1}{T_b} \int_{T_{\text{burn}}}^{T_{\text{burn}}+T_b} (\varphi(t) - \hat{\varphi}(t))^2 dt \right)^{1/2}. \quad (122)$$

The Kalman–Bucy filter is used here as a benchmark estimator. The main question of the chapter is not only how to run the filter, but why the filtering performance changes across regimes. This motivates the structural observability analysis below.

30. Finite-horizon observability and phase visibility

Throughout the chapter, we work in the classical linear-Gaussian framework. This is justified by the standard reduction of the Belavkin filter to the Kalman–Bucy filter for Gaussian linear quantum systems observed by homodyne detection [7, 3, 6]. We first ignore stochastic forcing and study how an initial perturbation in the phase direction propagates to the measurement channel. Consider the deterministic system

$$\dot{X}_a(t) = A_a X_a(t), \quad y(t) = C_a X_a(t), \quad (123)$$

where $y(t)$ is the noise-free output associated with the homodyne projection. The finite-horizon observability Gramian over $[0, T]$ is

$$W_o(T) = \int_0^T e^{A_a^\top t} C_a^\top C_a e^{A_a t} dt. \quad (124)$$

The pair (A_a, C_a) is observable over the horizon if this Gramian is positive definite. However, full-state observability is not the only relevant question here. The specific target is the phase component. We therefore define the directional phase-observability metric

$$J_\varphi(T) = e_\varphi^\top W_o(T) e_\varphi = \int_0^T \|C_a e^{A_a t} e_\varphi\|^2 dt. \quad (125)$$

The quantity $J_\varphi(T)$ is large when an initial perturbation in φ produces a large output signal over $[0, T]$, and small when the phase direction is nearly invisible from the measurement.

31. Resonant phase-visibility condition

A particularly transparent case is the resonant case

$$\omega = 0.$$

Then the augmented drift and measurement matrices are

$$A_a = \begin{bmatrix} -\gamma/2 & 0 & 0 \\ 0 & -\gamma/2 & g_\varphi \\ 0 & 0 & -\lambda \end{bmatrix}, \quad C_a = \sqrt{\kappa_m} \begin{bmatrix} \cos \theta & \sin \theta & 0 \end{bmatrix}. \quad (126)$$

The phase basis vector is $e_\varphi = [0 \ 0 \ 1]^\top$. Since $C_a e_\varphi = 0$, the phase does not appear directly in the measured output. We must look at how it enters through the dynamics:

$$A_a e_\varphi = \begin{bmatrix} 0 \\ g_\varphi \\ -\lambda \end{bmatrix}.$$

Projecting this propagated direction onto the homodyne output gives

$$C_a A_a e_\varphi = \sqrt{\kappa_m} g_\varphi \sin \theta. \quad (127)$$

Proposition 1. *In the resonant case $\omega = 0$, the hidden phase can influence the output trajectory at first order only if*

$$g_\varphi \sin \theta \neq 0. \quad (128)$$

Proof. (More Details in Annexe) For $\omega = 0$, the noise-free output is

$$y(t) = C_a X_a(t) = \sqrt{\kappa_m} (q(t) \cos \theta + p(t) \sin \theta).$$

The phase is not directly present in $y(t)$, since $C_a e_\varphi = 0$. The first place where it can appear is the output derivative:

$$\dot{y}(t) = C_a \dot{X}_a(t) = C_a A_a X_a(t).$$

Using (126), one obtains

$$C_a A_a = \sqrt{\kappa_m} \begin{bmatrix} -\frac{\gamma}{2} \cos \theta & -\frac{\gamma}{2} \sin \theta & g_\varphi \sin \theta \end{bmatrix}.$$

Therefore,

$$\dot{y}(t) = -\sqrt{\kappa_m} \frac{\gamma}{2} \cos \theta q(t) - \sqrt{\kappa_m} \frac{\gamma}{2} \sin \theta p(t) + \sqrt{\kappa_m} g_\varphi \sin \theta \varphi(t).$$

The coefficient multiplying $\varphi(t)$ is $\sqrt{\kappa_m} g_\varphi \sin \theta$. Since $\kappa_m > 0$, it is nonzero if and only if $g_\varphi \sin \theta \neq 0$. $\square$

The physical meaning is simple. The hidden phase must first excite the cavity through $g_\varphi \neq 0$, and the homodyne detector must project onto a quadrature that contains this phase information. In the resonant convention used here, the phase enters the p -quadrature. Therefore, geometries near $\theta = \pi/2$, where the homodyne detector measures mostly $p(t)$, are favorable. By contrast, $\theta \simeq 0$ measures mostly $q(t)$, which is blind to the phase in the resonant model.

32. Output-energy interpretation of $J_\varphi(T)$

The metric $J_\varphi(T)$ has an exact deterministic energy interpretation. Consider the system (123) with initial condition restricted to the phase direction:

$$X_a(0) = \alpha_0 e_\varphi, \quad \alpha_0 \in \mathbb{R}.$$

The scalar α_0 is the initial amplitude of the perturbation along the hidden phase coordinate; it should not be confused with the steady-state cavity amplitude α of Eq. (62), which is a complex number characterizing the coherent operating point of the cavity. Throughout this section, α_0 is a real scalar parametrizing the magnitude of a deterministic perturbation in the phase direction. The state and output trajectories are then

$$X_a(t) = \alpha_0 e^{A_a t} e_\varphi, \quad y(t) = \alpha_0 C_a e^{A_a t} e_\varphi.$$

Therefore,

$$\int_0^T \|y(t)\|^2 dt = \alpha_0^2 \int_0^T \|C_a e^{A_a t} e_\varphi\|^2 dt = \alpha_0^2 J_\varphi(T). \quad (129)$$

Lemma 1. *For the deterministic system $\dot{X}_a = A_a X_a$, $y = C_a X_a$, with initial condition $X_a(0) = \alpha_0 e_\varphi$, the output energy over $[0, T]$ satisfies*

$$\int_0^T \|y(t)\|^2 dt = \alpha_0^2 J_\varphi(T).$$

Moreover,

$$J_\varphi(T) = 0 \iff C_a e^{A_a t} e_\varphi = 0 \text{ for all } t \in [0, T].$$

Proof. The first identity follows from the explicit solution formula. The equivalence follows because the integrand $\|C_a e^{A_a t} e_\varphi\|^2$ is nonnegative. $\square$

This lemma shows that $J_\varphi(T)$ is not an abstract algebraic number only extracted from the Gramian. It is exactly the output energy generated by a unit initial perturbation ($\alpha_0 = 1$) in the hidden phase direction.

33. Directional finite-horizon estimation bound

The same quantity also has a measurement-side estimation interpretation. Consider the deterministic initial-condition problem

$$X_a(0) = \alpha_0 e_\varphi,$$

with $\alpha_0 \in \mathbb{R}$ the (unknown) phase-direction perturbation amplitude introduced in Section 32., observed through the noisy continuous-time measurement

$$dY(t) = \alpha_0 C_a e^{A_a t} e_\varphi dt + \sigma_v dV(t). \quad (130)$$

The unknown scalar is α_0 . This is a scalar continuous-time linear-Gaussian estimation problem. The Fisher information contained in the record over $[0, T]$ is

$$\mathcal{I}_T(\alpha_0) = \frac{1}{\sigma_v^2} \int_0^T \|C_a e^{A_a t} e_\varphi\|^2 dt = \frac{J_\varphi(T)}{\sigma_v^2}.$$

The classical Cramér–Rao inequality then gives

$$\text{Var}(\hat{\alpha}_0 - \alpha_0) \geq \frac{\sigma_v^2}{J_\varphi(T)} \quad (131)$$

for any unbiased estimator $\hat{\alpha}_0$ of α_0 .

Proposition 2. *In the restricted deterministic problem $X_a(0) = \alpha_0 e_\varphi$, observed through Eq. (130), any unbiased estimator $\hat{\alpha}_0$ of α_0 satisfies*

$$\text{Var}(\hat{\alpha}_0 - \alpha_0) \geq \frac{\sigma_v^2}{J_\varphi(T)}.$$

Proof. The result follows directly from the Fisher-information computation above. $\square$

This proposition should not be overinterpreted. It is not a full quantum Cramér–Rao bound, and it is not a complete bound for the stochastic Ornstein–Uhlenbeck filtering problem. It isolates only the measurement-side information associated with the phase-direction amplitude α_0 in a deterministic initial-condition setting. Full waveform estimation and continuously observed quantum systems involve additional considerations [17, 19, 20, 36]. Here, the result is used as an interpretation of directional visibility.

34. Steady-state phase variance and non-collapse

The metric $J_\varphi(T)$ captures a structural phase-to-output transfer. However, the Kalman filtering error depends not only on this deterministic visibility, but also on the stochastic excitation of the phase and on the measurement-noise level. This explains why one should not expect a universal one-parameter collapse of RMSE as a function of $J_\varphi(T)$. We make this point explicit in the resonant case $\omega = 0$. Let P_∞ denote the steady-state Kalman–Bucy error covariance, assuming the Riccati equation converges. It is the positive semidefinite solution of the algebraic Riccati equation

$$A_a P_\infty + P_\infty A_a^\top + Q - P_\infty C_a^\top R^{-1} C_a P_\infty = 0. \quad (132)$$

We write its entries as

$$P_\infty = (p_{ij})_{1 \leq i, j \leq 3}.$$

The phase-estimation error variance is the $(3, 3)$ entry:

$$[P_\infty]_{\varphi\varphi} = p_{33}.$$

The covariance between the measured homodyne direction and the hidden phase is

$$p_{\theta\varphi} = \cos \theta p_{13} + \sin \theta p_{23}. \quad (133)$$

The term p_{13} is the covariance between q and φ , while p_{23} is the covariance between p and φ .

Proposition 3. *In the resonant case $\omega = 0$, the steady-state phase-error variance satisfies*

$$[P_\infty]_{\varphi\varphi} = \frac{q_\varphi}{2\lambda} - \frac{\kappa_m p_{\theta\varphi}^2}{2\lambda \sigma_v^2}. \quad (134)$$

Proof. Extract the $(3, 3)$ entry of (132). Since the third row of A_a is $[0 \ 0 \ -\lambda]$, one has

$$(A_a P_\infty)_{33} = -\lambda p_{33}, \quad (P_\infty A_a^\top)_{33} = -\lambda p_{33}.$$

The process covariance contribution is

$$Q_{33} = q_\varphi.$$

Moreover,

$$C_a = \sqrt{\kappa_m} \begin{bmatrix} \cos \theta & \sin \theta & 0 \end{bmatrix}.$$

Therefore, the third component of $C_a P_\infty$ is

$$(C_a P_\infty)_3 = \sqrt{\kappa_m} (\cos \theta p_{13} + \sin \theta p_{23}) = \sqrt{\kappa_m} p_{\theta\varphi}.$$

Since $R = \sigma_v^2$,

$$(P_\infty C_a^\top R^{-1} C_a P_\infty)_{33} = \frac{\kappa_m p_{\theta\varphi}^2}{\sigma_v^2}.$$

Substituting these terms into the $(3, 3)$ entry of (132) gives

$$-2\lambda p_{33} + q_\varphi - \frac{\kappa_m p_{\theta\varphi}^2}{\sigma_v^2} = 0,$$

which is equivalent to (134). □

Equation (134) separates two effects. The term

$$\frac{q_\varphi}{2\lambda}$$

is the intrinsic stationary variance of the Ornstein–Uhlenbeck phase. The second term is the reduction due to measurement information. This reduction is controlled not only by the homodyne gain κ_m and noise level σ_v^2 , but also by the closed-loop covariance $p_{\theta\varphi}$, which is determined implicitly by the full Riccati equation. Thus, the phase variance is not governed by directional visibility alone. At the favorable homodyne angle $\theta = \pi/2$, one has $\cos \theta = 0$ and $\sin \theta = 1$, so

$$p_{\theta\varphi} = p_{23}.$$

The variance formula becomes

$$[P_\infty]_{\varphi\varphi} \Big|_{\theta=\pi/2} = \frac{q_\varphi}{2\lambda} - \frac{\kappa_m p_{23}^2}{2\lambda \sigma_v^2}. \quad (135)$$

This form is directly connected to the fact that, in the resonant model, the hidden phase enters the p -quadrature.

34.1 Closed form of $J_\varphi(\infty)$

The infinite-horizon directional metric can also be computed explicitly in the resonant case. Let

$$\beta = \frac{\gamma}{2}.$$

Starting from $X_a(0) = e_\varphi$, the phase trajectory is

$$\varphi(t) = e^{-\lambda t}.$$

The p -quadrature satisfies

$$\dot{p}(t) = -\beta p(t) + g_\varphi e^{-\lambda t}, \quad p(0) = 0.$$

If $\lambda \neq \beta$, then

$$p(t) = g_\varphi \frac{e^{-\lambda t} - e^{-\beta t}}{\beta - \lambda}.$$

The measured output generated by this initial phase perturbation is

$$y(t) = \sqrt{\kappa_m} \sin \theta p(t),$$

because the q -component is not driven in the resonant phase-only trajectory. Substituting into the definition of $J_\varphi(\infty)$ gives

$$J_\varphi(\infty) = \frac{g_\varphi^2 \kappa_m \sin^2 \theta}{2\lambda \left(\frac{\gamma}{2}\right) \left(\lambda + \frac{\gamma}{2}\right)}. \quad (136)$$

The same expression is obtained by continuity when $\lambda = \gamma/2$. This formula is useful because it displays the structural dependencies clearly:

$$J_\varphi(\infty) \propto g_\varphi^2 \kappa_m \sin^2 \theta.$$

Thus, phase visibility increases with stronger phase-to-quadrature coupling, stronger homodyne gain, and better alignment with the phase-sensitive quadrature. It decreases when the phase decays too quickly or when cavity damping suppresses the build-up of phase-dependent quadrature motion. However, comparing (136) with (134) shows why a universal scalar collapse is not expected. The expression for $J_\varphi(\infty)$ is an open-loop directional output-energy metric. The expression for $[P_\infty]_{\varphi\varphi}$ contains the intrinsic OU variance $q_\varphi/(2\lambda)$ and the Riccati-dependent covariance $p_{\theta\varphi}$. Two parameter sets may therefore have similar $J_\varphi(\infty)$ but different steady-state filtering errors because their stochastic phase renewal and covariance balance are different.

35. Augmented preprocessing cavity

We now introduce an auxiliary cavity coupled to the sensing cavity before measurement. The aim is not to build a full physical quantum network at this stage. The aim is to study a reduced linear model of dynamical preprocessing: additional dynamics are inserted between the phase-sensitive cavity and the homodyne readout, and we ask whether they improve or degrade phase estimation. The sensing cavity is indexed by s , and the auxiliary cavity by o . The sensing-cavity quadrature state is

$$x_s(t) = \begin{bmatrix} q_s(t) \\ p_s(t) \end{bmatrix},$$

where $q_s(t)$ and $p_s(t)$ are the sensing-cavity quadratures. The auxiliary-cavity quadrature state is

$$x_o(t) = \begin{bmatrix} q_o(t) \\ p_o(t) \end{bmatrix},$$

where $q_o(t)$ and $p_o(t)$ are the auxiliary quadratures. The full augmented state is

$$X_Q(t) = \begin{bmatrix} q_s(t) \\ p_s(t) \\ q_o(t) \\ p_o(t) \\ \varphi(t) \end{bmatrix} = \begin{bmatrix} x_s(t) \\ x_o(t) \\ \varphi(t) \end{bmatrix}. \quad (137)$$

The subscript Q indicates the two-cavity augmented model. The local drift matrices of the sensing and auxiliary cavities are

$$A_s = \begin{bmatrix} -\gamma_s/2 & \omega_s \\ -\omega_s & -\gamma_s/2 \end{bmatrix}, \quad A_o = \begin{bmatrix} -\gamma_o/2 & \omega_o \\ -\omega_o & -\gamma_o/2 \end{bmatrix}. \quad (138)$$

Here, $\gamma_s, \gamma_o > 0$ are the linewidths of the sensing and auxiliary cavities, while ω_s and ω_o are their detunings or quadrature-rotation frequencies in the rotating frame. The hidden phase drives only the sensing cavity, through

$$B_s = \begin{bmatrix} 0 \\ g_\varphi \end{bmatrix}. \quad (139)$$

The coupling from the auxiliary cavity to the sensing cavity is denoted by $K_{s \leftarrow o}$. The coupling from the sensing cavity to the auxiliary cavity is denoted by $K_{o \leftarrow s}$. This notation makes the direction of each coupling explicit. The coupled dynamics are

$$dx_s(t) = (A_s x_s(t) + K_{s \leftarrow o} x_o(t) + B_s \varphi(t)) dt + G_s dW_s(t), \quad (140)$$

$$dx_o(t) = (A_o x_o(t) + K_{o \leftarrow s} x_s(t)) dt + G_o dW_o(t). \quad (141)$$

The matrices G_s and G_o map vacuum-driven process noise into the sensing and auxiliary cavities. The processes $W_s(t)$ and $W_o(t)$ are independent two-dimensional Wiener processes. For numerical studies, we use the isotropic coupling surrogate

$$K_{s \leftarrow o} = k_{s \leftarrow o} I_2, \quad K_{o \leftarrow s} = k_{o \leftarrow s} I_2, \quad (142)$$

where $k_{s \leftarrow o}$ and $k_{o \leftarrow s}$ are scalar coupling strengths. This is a simplified state-space representation of linear quadrature-to-quadrature exchange. If one wants to impose a specific Hamiltonian beam-splitter interaction or exact physical realizability, then the coupling and noise matrices must satisfy the corresponding algebraic constraints for linear quantum stochastic systems [23, 34, 24, 25]. Here, the goal is more limited: to isolate the effect of auxiliary dynamics on phase visibility and filtering performance. Combining the sensing cavity, auxiliary cavity, and OU phase dynamics gives

$$dX_Q(t) = A_Q X_Q(t) dt + G_Q dW_Q(t), \quad (143)$$

where

$$A_Q = \begin{bmatrix} A_s & K_{s \leftarrow o} & B_s \\ K_{o \leftarrow s} & A_o & 0 \\ 0 & 0 & -\lambda \end{bmatrix}. \quad (144)$$

With the isotropic coupling (142), this matrix is

$$A_Q = \begin{bmatrix} -\gamma_s/2 & \omega_s & k_{s \leftarrow o} & 0 & 0 \\ -\omega_s & -\gamma_s/2 & 0 & k_{s \leftarrow o} & g_\varphi \\ k_{o \leftarrow s} & 0 & -\gamma_o/2 & \omega_o & 0 \\ 0 & k_{o \leftarrow s} & -\omega_o & -\gamma_o/2 & 0 \\ 0 & 0 & 0 & 0 & -\lambda \end{bmatrix}. \quad (145)$$

The augmented noise process is

$$W_Q(t) = \begin{bmatrix} W_s(t) \\ W_o(t) \\ W_\varphi(t) \end{bmatrix},$$

and the noise matrix is block diagonal:

$$G_Q = \text{diag} \left(G_s, G_o, \sqrt{q_\varphi} \right). \quad (146)$$

In the vacuum-limited normalized case, one may take

$$G_s = \sqrt{\gamma_s} I_2, \quad G_o = \sqrt{\gamma_o} I_2.$$

36. Augmented homodyne readout

The two-cavity output is modeled as a linear homodyne projection of accessible quadratures. The measurement equation is

$$dY(t) = C_Q X_Q(t) dt + \sigma_v dV(t), \quad (147)$$

where C_Q is the augmented readout row. We write it as

$$C_Q = \sqrt{\kappa_m} \begin{bmatrix} \ell_s \cos \theta_s & \ell_s \sin \theta_s & \ell_o \cos \theta_o & \ell_o \sin \theta_o & 0 \end{bmatrix}. \quad (148)$$

Here, θ_s is the quadrature angle associated with the sensing-cavity output, θ_o is the quadrature angle associated with the auxiliary output, and $\ell_s, \ell_o \in \mathbb{R}$ are readout weights. If the two outputs are mixed in a normalized lossless way, one may impose

$$\ell_s^2 + \ell_o^2 = 1.$$

If only the sensing-cavity output is measured, then $\ell_s = 1$ and $\ell_o = 0$. If the auxiliary output also contributes to the homodyne signal, then $\ell_o \neq 0$. The measured deterministic output is therefore

$$C_Q X_Q(t) = \sqrt{\kappa_m} \left[\ell_s (q_s(t) \cos \theta_s + p_s(t) \sin \theta_s) + \ell_o (q_o(t) \cos \theta_o + p_o(t) \sin \theta_o) \right]. \quad (149)$$

Again, the phase is not directly measured. If

$$e_{\varphi, Q} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

then

$$C_Q e_{\varphi, Q} = 0. \quad (150)$$

The phase must therefore pass through the sensing cavity, then possibly through the auxiliary dynamics, and finally through the homodyne projection:

$$\varphi(t) \longrightarrow p_s(t) \longrightarrow x_s(t) \longleftrightarrow x_o(t) \longrightarrow Y(t).$$

The complete augmented preprocessing model is

$$\boxed{\begin{aligned} dX_Q(t) &= A_Q X_Q(t) dt + G_Q dW_Q(t), \\ dY(t) &= C_Q X_Q(t) dt + \sigma_v dV(t). \end{aligned}} \quad (151)$$

37. Visibility and noise trade-off

The auxiliary cavity changes the phase-to-output transfer, but it also changes the stochastic structure of the estimation problem. Therefore, its benefit cannot be measured only by the presence of additional dynamics. It must be evaluated through both information transfer and noise propagation. For the plant-only model, the directional phase metric is

$$J_\varphi^{\text{plant}}(T) = e_\varphi^\top W_o^{\text{plant}}(T) e_\varphi,$$

where $W_o^{\text{plant}}(T)$ is computed from the pair (A_a, C_a) . For the augmented model, the corresponding metric is

$$J_\varphi^{\text{aug}}(T) = e_{\varphi,Q}^\top W_o^{\text{aug}}(T) e_{\varphi,Q},$$

where $W_o^{\text{aug}}(T)$ is computed from the pair (A_Q, C_Q) . To compare both architectures, we define the relative visibility gain

$$\Delta_{J_\varphi} = \frac{J_\varphi^{\text{aug}}(T) - J_\varphi^{\text{plant}}(T)}{\max(J_\varphi^{\text{plant}}(T), \varepsilon_{\text{reg}})}. \quad (152)$$

The constant $\varepsilon_{\text{reg}} > 0$ is a small regularization constant introduced only to avoid division by a very small denominator. A positive value of Δ_{J_φ} means that the augmented dynamics increase directional phase visibility. To quantify the stochastic cost in the numerical study, we also use a relative diffusion variation:

$$\Delta_Q = \frac{\text{tr}(Q_d^{\text{aug}}) - \text{tr}(Q_d^{\text{plant}})}{\max(\text{tr}(Q_d^{\text{plant}}), \varepsilon_{\text{reg}})}. \quad (153)$$

Here, Q_d^{plant} and Q_d^{aug} are the discrete-time process covariances used in simulation after discretizing the continuous-time models. This Δ_Q is not part of the continuous-time observability theory. It is only an empirical proxy for the additional stochastic burden seen by the numerical estimator. Finally, the relative phase-estimation improvement is defined as

$$\Delta_{\text{RMSE}} = \frac{\text{RMSE}_{\text{plant}} - \text{RMSE}_{\text{aug}}}{\text{RMSE}_{\text{plant}}}. \quad (154)$$

Thus, $\Delta_{\text{RMSE}} > 0$ means that the augmented architecture improves phase estimation, while $\Delta_{\text{RMSE}} < 0$ means that it degrades performance.

38. Numerical study

The numerical simulations compare the plant-only model (117) and the augmented preprocessing model (151). For each operating point, the continuous-time dynamics are discretized for simulation, and the hidden phase is estimated using a discrete-time Kalman filter consistent with the continuous-time Kalman–Bucy model [27, 28, 29]. The main sweep is performed over two parameters:

$$\theta, \quad \lambda.$$

The angle θ controls the homodyne measurement geometry. The rate λ controls the renewal speed of the Ornstein–Uhlenbeck phase process. For each pair (θ, λ) , the phase RMSE is computed after a burn-in period and averaged over N_{MC} independent Monte Carlo runs. The nominal parameters used in the (θ, λ) sweep are

$$\begin{aligned} \gamma_s &= 1.0, & \omega_s &= 0, & \gamma_o &= 1.2, & \omega_o &= 0.4, \\ g_\varphi &= 2.0, & \kappa_m &= 1.0, & q_\varphi &= 0.01, & \sigma_v &= 0.03. \end{aligned}$$

The coupling strengths in the augmented model are chosen as

$$k_{s \leftarrow o} = k_{o \leftarrow s} = 0.8.$$

The simulation horizon is

$$T = 60.0,$$

the numerical time step is

$$\Delta t = 0.015,$$

and the burn-in fraction is 20% of the trajectory. The homodyne angle is sampled uniformly over

$$[0.05, \pi - 0.05],$$

and the OU rate is sampled uniformly over

$$[0.02, 1.5].$$

The grid size is 64×48 , giving 3072 operating points. For each operating point, $N_{\text{MC}} = 20$ Monte Carlo trajectories are used. Random realizations are generated with a deterministic global seed to ensure reproducibility.

38.1 Plant-only visibility trends

We first examine the plant-only architecture. Figure 4 shows RMSE families at fixed OU rate λ as a function of $1/\sqrt{J_\varphi(T)}$, with the homodyne angle θ varying along each curve. Within each fixed- λ family, the RMSE increases as $J_\varphi(T)$ decreases. This is consistent with Proposition 1: when the measurement angle moves away from the favorable phase-quadrature geometry near $\theta = \pi/2$, the phase becomes less visible and estimation becomes harder. This local predictive value

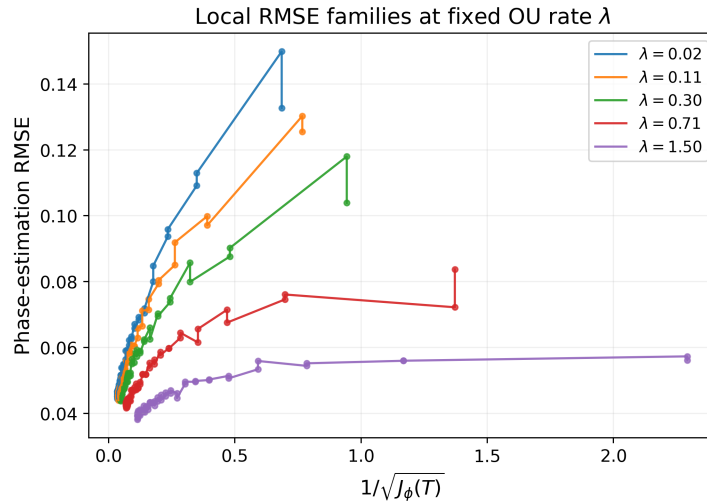


Figure 4: Plant-only RMSE families at fixed OU rate λ , plotted as a function of $1/\sqrt{J_\varphi(T)}$. Along each curve, the homodyne angle θ varies. The monotone behavior within each family shows that $J_\varphi(T)$ is an informative intra-family visibility metric.

does not extend globally across mixed regimes. Figure 5 shows RMSE versus $J_\varphi(T)$ across the full (θ, λ) sweep, with points colored by λ . If $J_\varphi(T)$ were a sufficient universal predictor, the point cloud would collapse to a single curve. Instead, a clear stratification remains. This confirms that λ acts as an additional regime parameter beyond directional visibility alone. We now compare the plant-only and augmented architectures. Figure 6 shows the relative RMSE improvement Δ_{RMSE} over the (θ, λ) plane. Positive values correspond to improved estimation due to the auxiliary cavity. Negative values correspond to degradation.

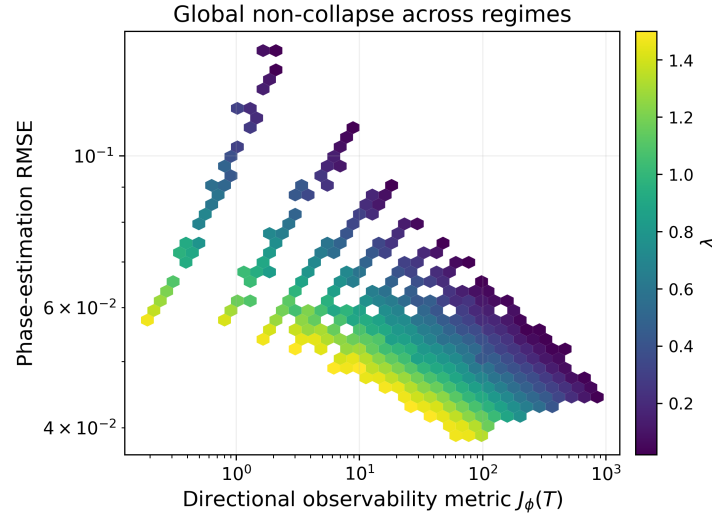


Figure 5: Global non-collapse of RMSE versus $J_\varphi(T)$ across the full (θ, λ) sweep. Points are colored by the OU rate λ . The remaining stratification shows that phase renewal is an independent stochastic regime parameter.

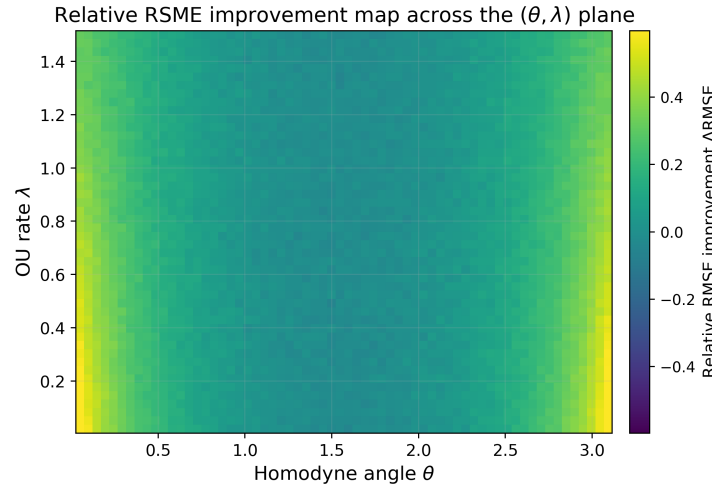


Figure 6: Relative RMSE improvement of the augmented architecture over the plant-only baseline:

$$\Delta_{\text{RMSE}} = \frac{\text{RMSE}_{\text{plant}} - \text{RMSE}_{\text{aug}}}{\text{RMSE}_{\text{plant}}}.$$

Positive values indicate benefit from augmented preprocessing, while negative values indicate degradation. The map shows that the auxiliary cavity is beneficial only in part of the parameter space.

The map shows that preprocessing is not uniformly beneficial. It helps in regimes where the auxiliary dynamics increase useful phase visibility without too much stochastic penalty. It can hurt near blind angles or in regimes where the additional dynamics mainly add noise or create weakly observable modes.

Figure 7 compares Δ_{RMSE} against the relative visibility gain Δ_{J_φ} . A positive trend appears: configurations with larger visibility gain tend to produce larger RMSE improvement. This supports the interpretation that, in geometry-dominated regimes, the auxiliary cavity helps by increasing directional phase visibility. The remaining spread at fixed Δ_{J_φ} also shows that visibility gain alone does not explain all performance variations.

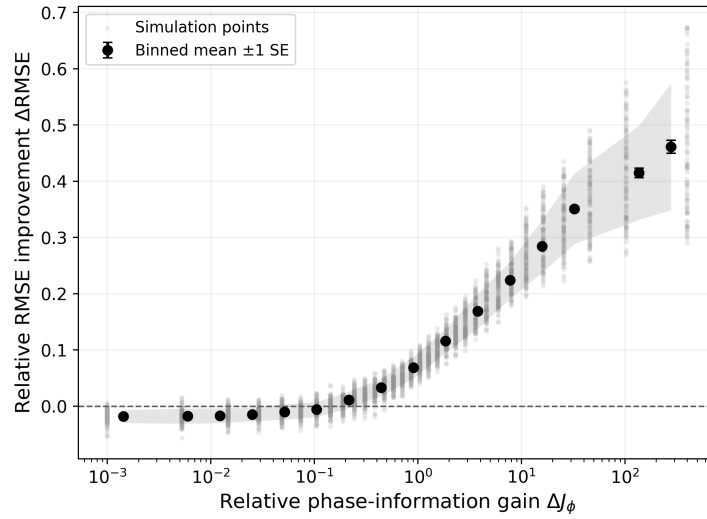
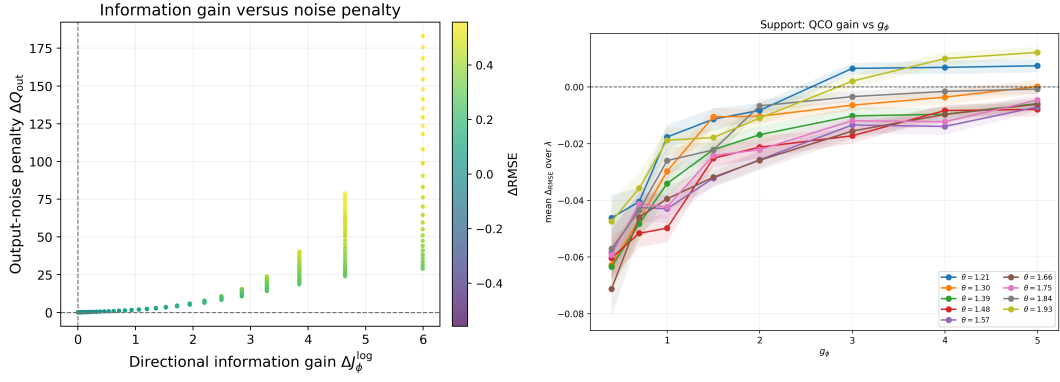


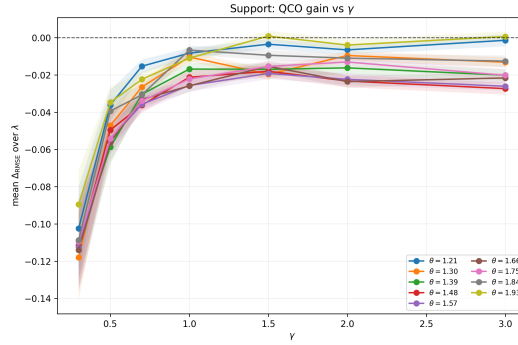
Figure 7: Relative RMSE improvement of the augmented architecture versus relative visibility gain ΔJ_φ . The positive trend supports a visibility-enhancement interpretation, while the residual spread shows that noise propagation and dynamical effects also matter.

38.2 Support plots: visibility gain versus noise burden

To separate the competing mechanisms more clearly, Figure 8 reports three support plots. The first compares a directional information gain with an output-noise penalty, while the color scale indicates Δ_{RMSE} . The second shows the mean Δ_{RMSE} over λ as a function of the phase-coupling strength g_φ , for several homodyne angles. The third shows the same averaged improvement as a function of the cavity linewidth γ . The first panel illustrates the central trade-off. Configurations with stronger phase-information transfer often also carry a larger noise penalty. Therefore, the auxiliary cavity should not be interpreted as a universally beneficial correction. It reshapes both the useful signal pathway and the effective noise seen by the estimator. The second panel shows that when g_φ is small, the augmented architecture is generally detrimental. In that regime, the hidden phase is only weakly imprinted onto the sensing cavity, so the auxiliary subsystem has little useful phase information to redistribute. As g_φ increases, the preprocessing penalty decreases, and for favorable homodyne angles it can turn into a positive gain. The third panel shows the role of the linewidth γ . For very small γ , the augmented architecture suffers a strong degradation. As γ increases toward moderate values, the penalty decreases and the curves approach a balanced regime. This suggests that excessively weak damping makes the additional dynamics mostly costly, whereas moderate damping allows the auxiliary cavity to reshape the phase-to-output pathway more effectively.



(a) Directional information gain versus output-noise penalty, colored by Δ_{RMSE} . (b) Mean Δ_{RMSE} over λ as a function of g_φ , for several homodyne angles θ .



(c) Mean Δ_{RMSE} over λ as a function of γ , for several homodyne angles θ .

Figure 8: Support plots for the augmented-architecture trade-off. The auxiliary subsystem can improve phase-direction transfer, but this benefit must be balanced against the additional stochastic burden introduced by the extra dynamics.

39. Conclusion of the chapter

The results of this chapter support three conclusions. First, phase estimation in the linearized cavity model is strongly constrained by observability geometry. In the resonant case, the phase enters the p -quadrature, so homodyne angles close to $\theta = \pi/2$ are favorable, whereas angles close to $\theta = 0$ are nearly blind to the phase channel. Second, the directional metric $J_\varphi(T)$ is meaningful but not universal. It gives an exact output-energy measure of the phase direction and explains RMSE trends within fixed stochastic regimes. However, it does not produce a global one-parameter collapse across mixed regimes, because the OU relaxation rate, phase diffusion intensity, measurement noise, and Riccati covariance structure all affect the filtering error. Third, augmented preprocessing through an auxiliary cavity is intrinsically regime-dependent. It can improve estimation by increasing phase-direction visibility, but it can also degrade estimation by adding noise or producing unfavorable dynamical redistribution. The correct interpretation is therefore a visibility–noise trade-off, not a universal advantage of the augmented architecture.

Chapitre 6 – Classical Extended Kalman Filtering for phase parameter estimation

40. Extended Kalman Filter for nonlinear phase estimation

In the linearized cavity models of the chapter on the Optical cavity formulation as a phase-estimation problem and the chapter on Cavity Phase Estimation: Observability Limits and Regime-Dependent Preprocessing Trade-Offs, the homodyne signal was an affine function of the augmented state, and a linear Kalman–Bucy filter was sufficient. In the present chapter, we drop the small-phase approximation $e^{i\varphi} \simeq 1 + i\varphi$ used in the chapter on the Optical cavity formulation as a phase-estimation problem: the phase enters the dynamics and the homodyne signal through the trigonometric terms $\cos \varphi$ and $\sin \varphi$. The standard linear Kalman filter is therefore replaced by an extended Kalman filter (EKF), which locally linearizes the nonlinear state and measurement maps around the current estimate [29, 28]. For consistency with the chapter on Homodyne Detection, the chapter on the Optical cavity formulation as a phase-estimation problem and the chapter on Cavity Phase Estimation: Observability Limits and Regime-Dependent Preprocessing Trade-Offs, the hidden phase is denoted by $\varphi(t)$. The phase-diffusion intensity used in this chapter is written σ_φ^2 ; it is related to the phase-diffusion parameter q_φ of the chapter on the Classical Observer for Phase Estimation via Homodyne Detection through $q_\varphi = \sigma_\varphi^2$, so that the continuous-time OU model $d\varphi = -\lambda\varphi dt + \sqrt{q_\varphi} dW_\varphi$ and its discrete-time counterpart (158) below describe the same process. The cavity linewidths are denoted by $\gamma_s, \gamma_o > 0$, in agreement with the symbol γ used for the single-cavity linewidth in the chapter on the Optical cavity formulation as a phase-estimation problem; the symbol κ_m used elsewhere refers to the homodyne measurement gain and is distinct from the cavity linewidths. Throughout this chapter, $A_0 > 0$ denotes the magnitude of the steady coherent operating amplitude introduced in the chapter on the Optical cavity formulation as a phase-estimation problem: $A_0 = |\alpha|$, where

$$\alpha = \frac{\varepsilon}{\frac{\gamma_s}{2} + i\omega_s}$$

is the steady-state intracavity amplitude defined in Eq. (62). Equivalently, A_0 coincides with the real signal amplitude A_s of the chapter on Homodyne Detection and with the calibrated coherent amplitude $|\alpha|$ of the chapter on the Classical Observer for Phase Estimation via Homodyne Detection.

41. Augmented state and nonlinear discrete-time model

We consider the augmented state

$$X_k = \begin{bmatrix} q_{s,k} \\ p_{s,k} \\ q_{o,k} \\ p_{o,k} \\ \varphi_k \end{bmatrix}, \quad w_k \sim \mathcal{N}(0, Q), \quad v_k \sim \mathcal{N}(0, R). \quad (155)$$

The sensing cavity and the auxiliary cavity are described by the matrices

$$A_s = \begin{bmatrix} -\gamma_s/2 & \omega_s \\ -\omega_s & -\gamma_s/2 \end{bmatrix}, \quad A_o = \begin{bmatrix} -\gamma_o/2 & \omega_o \\ -\omega_o & -\gamma_o/2 \end{bmatrix}, \quad (156)$$

and their Euler discretizations

$$F_s = I_2 + A_s dt, \quad F_o = I_2 + A_o dt. \quad (157)$$

The phase is modeled as a discrete Ornstein–Uhlenbeck process,

$$\varphi_{k+1} = (1 - \lambda dt) \varphi_k + \sigma_\varphi \sqrt{dt} \xi_k, \quad \xi_k \sim \mathcal{N}(0, 1), \quad (158)$$

where $\lambda > 0$ is the phase relaxation rate and $\sigma_\varphi > 0$ is the diffusion amplitude (so that $\sigma_\varphi^2 = q_\varphi$ of the chapter on the Classical Observer for Phase Estimation via Homodyne Detection). A physically important distinction must be emphasized between the model used here and the two-cavity model of the chapter on the Optical cavity formulation as a phase-estimation problem. In that chapter, the sensing and auxiliary cavities were coupled through a direct Hamiltonian (bidirectional) coupling, encoded by the matrices $K_{s \leftarrow o}$ and $K_{o \leftarrow s}$; this coupling preserves the canonical structure of a closed quadrature network and admits, under additional algebraic constraints, a physically realizable quantum-optical implementation [23, 24, 25]. In contrast, the present chapter uses a unidirectional cascade coupling: the output field of the sensing cavity is fed as input field to the auxiliary cavity, in the spirit of Gardiner’s cascade input–output formalism [32, 26, 4]. Concretely, the cascade relation

$$a_{\text{in}}^{(o)}(t) = a_{\text{in}}^{(s), \text{vac}}(t) - \sqrt{\gamma_s} a_s(t), \quad (159)$$

inserted into the Langevin equation of the auxiliary cavity, produces the explicit cross-term $-\sqrt{\gamma_o}\sqrt{\gamma_s} q_s$ (and similarly for p_s) in the auxiliary-cavity update. This is a non-reciprocal information flow: the sensing cavity influences the auxiliary cavity, but not the converse. The two architectures are therefore not equivalent, and the results obtained here should be compared with those of the chapter on Cavity Phase Estimation: Observability Limits and Regime-Dependent Preprocessing Trade-Offs with this caveat in mind. The full nonlinear state transition is then written as

$$X_{k+1} = f(X_k) + w_k, \quad (160)$$

with

$$f(X) = \begin{bmatrix} F_s \begin{bmatrix} q_s \\ p_s \end{bmatrix} + dt \sqrt{2\gamma_s} A_o \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix} \\ F_o \begin{bmatrix} q_o \\ p_o \end{bmatrix} + dt \sqrt{\gamma_o} \begin{bmatrix} \sqrt{2} A_o \cos \varphi - \sqrt{\gamma_s} q_s \\ \sqrt{2} A_o \sin \varphi - \sqrt{\gamma_s} p_s \end{bmatrix} \\ (1 - \lambda dt) \varphi \end{bmatrix}. \quad (161)$$

The bracketed term $\sqrt{2} A_o \cos \varphi - \sqrt{\gamma_s} q_s$ (and its sine counterpart) is exactly the output q -quadrature (resp. p -quadrature) of the sensing cavity, in agreement with the cascade relation (159).

42. Homodyne measurement model

We define the intermediate output quadratures of the sensing cavity as

$$Q_{\text{out}}^{(s)}(X) = \sqrt{2} A_o \cos \varphi - \sqrt{\gamma_s} q_s, \quad P_{\text{out}}^{(s)}(X) = \sqrt{2} A_o \sin \varphi - \sqrt{\gamma_s} p_s, \quad (162)$$

and those of the auxiliary cavity as

$$Q_{\text{out}}^{(o)}(X) = Q_{\text{out}}^{(s)}(X) - \sqrt{\gamma_o} q_o, \quad P_{\text{out}}^{(o)}(X) = P_{\text{out}}^{(s)}(X) - \sqrt{\gamma_o} p_o. \quad (163)$$

The homodyne measurement is then

$$y_k = h(X_k, \theta_k) + v_k, \quad (164)$$

with

$$h(X, \theta) = \cos \theta Q_{\text{out}}^{(o)}(X) + \sin \theta P_{\text{out}}^{(o)}(X) - \sqrt{2}A_0 \cos \theta. \quad (165)$$

In the nonlinear EKF setting we adopt the normalized convention $\kappa_m = \gamma_s$ (resp. $\kappa_m = \gamma_o$ for the auxiliary output channel), so that the homodyne sensitivity factors $\sqrt{\kappa_m}$ of the chapters on Homodyne Detection, the Optical cavity formulation as a phase-estimation problem, and Cavity Phase Estimation: Observability Limits and Regime-Dependent Preprocessing Trade-Offs reduce to $\sqrt{\gamma_s}$ and $\sqrt{\gamma_o}$ in the measurement Jacobian H_k of Eq. (208). Equivalently,

$$\begin{aligned} h(X, \theta) = & \cos \theta \left(\sqrt{2}A_0 \cos \varphi - \sqrt{\gamma_s}q_s - \sqrt{\gamma_o}q_o \right) \\ & + \sin \theta \left(\sqrt{2}A_0 \sin \varphi - \sqrt{\gamma_s}p_s - \sqrt{\gamma_o}p_o \right) - \sqrt{2}A_0 \cos \theta. \end{aligned} \quad (166)$$

The constant subtracted in (165) is the coherent offset (or DC bias) of the homodyne photocurrent. In the chosen drive convention, the input field carries a steady mean amplitude $\langle a_{\text{in}} \rangle = \sqrt{2}A_0/\sqrt{\gamma_s}$ (in the units used here), so that the noise-free homodyne signal in absence of phase modulation ($\varphi \equiv 0$, $q_s = p_s = q_o = p_o = 0$) reduces to $h_{\text{DC}} = \sqrt{2}A_0 \cos \theta$. This DC component is deterministic, known a priori once θ is set, and carries no information about φ . It is therefore subtracted from the raw photocurrent before processing, in agreement with the standard balanced homodyne calibration procedure of the chapter on Homodyne Detection [3, 4]. After this subtraction, $h(X, \theta)$ is a zero-mean nonlinear measurement function whose Jacobian is well-behaved for the EKF linearization.

43. Adaptive homodyne angle

At each time step, the local-oscillator phase is chosen from the predicted phase estimate:

$$\theta_k = \text{wrap} \left(\hat{\varphi}_{k|k-1} + \frac{\pi}{2} \right). \quad (167)$$

This choice aligns the measured quadrature with the direction of maximal local phase sensitivity: indeed, expanding $h(X, \theta)$ around $\varphi = \hat{\varphi}_{k|k-1}$ and choosing $\theta = \hat{\varphi}_{k|k-1} + \pi/2$ maximizes $|\partial h / \partial \varphi|$ at first order. This adaptive homodyne strategy was originally proposed by Wiseman [8, 9] in the context of optical phase estimation and has since become a standard ingredient of optimal-feedback continuous quantum measurement [10, 3].

44. EKF recursion

The EKF prediction step is

$$\hat{X}_{k|k-1} = f(\hat{X}_{k-1|k-1}), \quad (168)$$

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^\top + Q, \quad F_k = \left. \frac{\partial f}{\partial X} \right|_{\hat{X}_{k-1|k-1}}. \quad (169)$$

The predicted measurement is

$$\hat{y}_{k|k-1} = h(\hat{X}_{k|k-1}, \theta_k), \quad (170)$$

and the measurement Jacobian is

$$H_k = \left. \frac{\partial h}{\partial X} \right|_{(\hat{X}_{k|k-1}, \theta_k)}. \quad (171)$$

The update step is then

$$K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R)^{-1}, \quad (172)$$

$$\widehat{X}_{k|k} = \widehat{X}_{k|k-1} + K_k (y_k - \widehat{y}_{k|k-1}), \quad (173)$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1}. \quad (174)$$

The phase estimate of interest is the fifth component of the state estimate,

$$\widehat{\varphi}_k = (\widehat{X}_{k|k})_5. \quad (175)$$

The explicit expression of the Jacobians F_k and H_k used in the simulation is reported in the appendix section EKF Jacobians used in Chapter 5. The EKF is known to be a locally optimal estimator only: it is the optimal linear minimum mean-square estimator for the system linearized around the current estimate, and it offers no global convergence guarantee [29, 28, 37]. In the present model, the measurement nonlinearity $\cos \varphi, \sin \varphi$ is 2π -periodic and bounded, while φ itself can drift with finite probability over excursions exceeding $\pi/2$. Two known failure modes therefore become possible:

- If $\widehat{\varphi}_{k|k-1}$ and the true φ_k fall on opposite slopes of the cosine (i.e., differ by approximately π), the linearized Jacobian H_k carries the wrong sign, and the innovation drives the estimator further away rather than correcting it. This is sometimes called fringe hopping in optical phase-estimation literature [10, 38].
- The Riccati update (174) systematically underestimates the true uncertainty when the second-order term in the Taylor expansion of h is non-negligible. The filter becomes overconfident, which exacerbates the previous failure mode.

For the operating regime considered in Section 46. (small-amplitude OU phase, $|\varphi|$ typically below 2 rad, and large drive amplitude $A_0 = 100$ yielding a high effective signal-to-noise ratio), neither failure mode was observed in our simulations. However, in regimes with larger phase excursions or weaker drive, the EKF can be advantageously replaced by a derivative-free Unscented Kalman Filter (UKF), which propagates a deterministic set of sigma points through the exact nonlinearity and avoids the linearization bias [39, 40]. The UKF, or alternatively a particle filter [41], would be a natural extension of the present work for handling the cycle-slip regime.

45. System used in the EKF simulation

The numerical experiment considered in this section is based on a cascaded architecture composed of three elements:

1. a latent phase φ_k following a discrete Ornstein–Uhlenbeck law,
2. a sensing cavity with quadratures $(q_{s,k}, p_{s,k})$,
3. an auxiliary cavity with quadratures $(q_{o,k}, p_{o,k})$,

followed by homodyne detection with an adaptive local-oscillator phase. The role of the auxiliary cavity is to preprocess the optical field before measurement, while the EKF reconstructs the hidden phase from the noisy homodyne record. The state used in the EKF is

$$X_k = [q_{s,k} \quad p_{s,k} \quad q_{o,k} \quad p_{o,k} \quad \varphi_k]^\top,$$

so that the estimator jointly reconstructs the two cavity states and the hidden phase. The first cavity is directly driven by the trigonometric phase terms $\cos \varphi_k$ and $\sin \varphi_k$, while the second

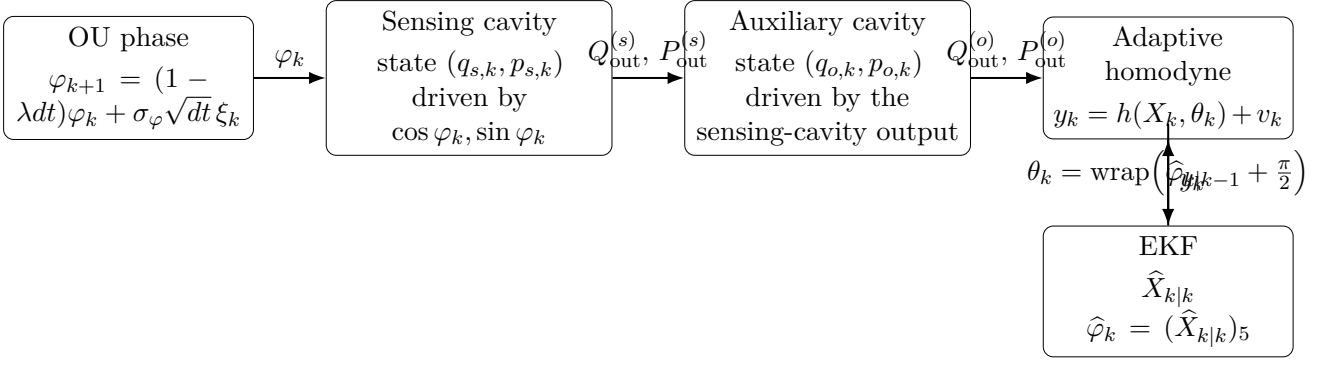


Figure 9: Architecture used in the nonlinear EKF simulation. A latent Ornstein–Uhlenbeck phase drives the sensing cavity, whose output feeds an auxiliary cavity through a unidirectional cascade coupling (159). The output of the auxiliary cavity is measured by adaptive homodyne detection, and the resulting noisy measurement is processed by an extended Kalman filter. The local-oscillator phase θ_k is fed back from the EKF prediction, following the adaptive scheme of Wiseman [8].

cavity receives as input the output quadratures of the first cavity through (159). This explains why the measurement depends nonlinearly on φ_k : the phase first modifies the field inside the sensing cavity, then propagates through the auxiliary cavity, and only finally appears in the measured homodyne quadrature. In the simulation reported here, the parameters were chosen as

$$dt = 0.01 \text{ s}, \quad T = 10 \text{ s}, \quad \gamma_s = \gamma_o = 1 \text{ s}^{-1}, \quad \omega_s = \omega_o = 0, \quad A_0 = 100, \quad R = 1,$$

with phase parameters $\lambda = 1 \text{ s}^{-1}$ and $\sigma_\varphi = 1 \text{ rad s}^{-1/2}$, and initial EKF conditions $\hat{X}_{0|0} = 0$ and $P_{0|0} = I_5$. These choices correspond exactly to the numerical script used to generate the tracking curves shown in the next section.

46. Results

This corresponds to a root-mean-square error $\text{RMSE} \approx 0.174 \text{ rad}$, i.e. about 10° , confirming quantitatively the tracking performance observed in Fig. 10. To isolate the contribution of the adaptive law (167), the same simulation was repeated with a constant local-oscillator phase $\theta_k \equiv \theta_0$, for $\theta_0 \in \{0, \pi/4, \pi/2\}$. The tail-averaged RMSE values are reported in Table 1.

Local-oscillator strategy	RMSE [rad]
Fixed $\theta = 0$ (q -quadrature)	≈ 0.43
Fixed $\theta = \pi/4$	≈ 0.27
Fixed $\theta = \pi/2$ (p -quadrature)	≈ 0.21
Adaptive $\theta_k = \hat{\varphi}_{k k-1} + \pi/2$	≈ 0.17

Table 1: Tail-averaged phase RMSE (last 20% of the trajectory) for the EKF combined with different homodyne strategies, on the cascaded two-cavity model of Section 45.. The adaptive strategy outperforms the best fixed angle by $\approx 20\%$. Single representative trajectory, fixed random seed.

The trend is consistent with the resonant phase-visibility condition of Chapter 5 (Proposition 1): $\theta_0 = 0$ measures the q -quadrature, which is structurally blind to the phase at first order in the resonant model; $\theta_0 = \pi/2$ measures the phase-sensitive p -quadrature and is much closer to the adaptive performance, but still loses information when the locally optimal angle drifts away from

$\pi/2$. The adaptive strategy tracks this drift, in agreement with the analysis of Wiseman [8, 9]. The adaptive law returns $\theta_k \in (-\pi, \pi]$ and therefore exhibits 2π -jumps at the branch boundary; these are an artifact of the representation, not of the dynamics, and have no effect on $h(X_k, \theta_k)$ (2π -periodic) nor on the EKF state estimate. The adaptive EKF reduces the tail-RMSE by $\approx 20\%$ over the best fixed-angle baseline in the small-excursion regime considered here. Its extension to large excursions, where cycle slips become probable, is naturally addressed by an unscented Kalman filter [39] or a particle filter [41], and is left for future work. The corresponding value is

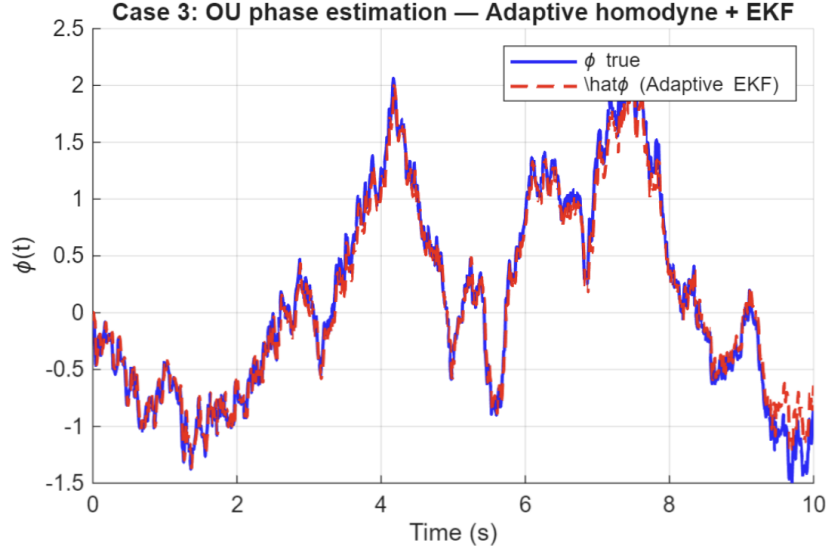


Figure 10: Adaptive EKF phase tracking for the nonlinear homodyne model. The blue solid curve is the true Ornstein–Uhlenbeck phase $\varphi(t)$, while the red dashed curve is the EKF estimate $\hat{\varphi}(t)$. The close agreement between the two curves shows that the adaptive homodyne strategy combined with the EKF is able to reconstruct the latent phase accurately in the nonlinear regime.

$$\text{MSE}_{80\%-100\%} = \frac{1}{N_{\text{tail}}} \sum_{k=\lfloor 0.8N \rfloor}^N (\varphi_k - \hat{\varphi}_k)^2 \approx 3.02 \times 10^{-2} \text{ rad}^2, \quad (176)$$

Chapitre 7 – Conclusion and perspectives

47. Summary of contributions

This report addressed the estimation of a hidden Ornstein–Uhlenbeck optical phase from balanced homodyne measurements, on architectures from a single cavity to two coupled cavities, in both the linear small-phase regime (Chapters 3 and 4) and the fully nonlinear regime (Chapter 5). In the linear regime, the cavity phase-estimation problem was formulated as a finite-horizon observability problem on an augmented state-space model. The directional metric $J_\varphi(T)$ was introduced and shown to be exactly the deterministic phase-direction output energy, yielding a Cramér–Rao-type lower bound. A closed-form steady-state phase-variance decomposition then clarified the role of $J_\varphi(T)$ as a faithful intra-family predictor that does not collapse the RMSE universally, since the OU rate, the phase-diffusion intensity, the measurement noise and the closed-loop covariance enter independently. A Hamiltonian-coupled two-cavity preprocessing architecture was studied as a regime-dependent visibility/noise trade-off, and the numerical sweep identified the operating regions where the auxiliary cavity actually helps. In the nonlinear regime, an extended Kalman filter combined with the adaptive law $\theta_k = \text{wrap}(\hat{\varphi}_{k|k-1} + \pi/2)$ reduced the tail-RMSE by $\approx 20\%$ over the best fixed-angle baseline, in agreement with the structural prediction that the phase enters the p -quadrature in the resonant model.

48. Limitations

Four limitations should be made explicit: (i) the linear models of Chapters 3 and 4 rely on the small-phase expansion $e^{i\varphi} \simeq 1 + i\varphi$, which breaks down for large excursions or low SNR where cycle slips become probable; (ii) the Hamiltonian coupling between the two cavities is modeled by an isotropic surrogate $K_{s \leftarrow o} = k_{s \leftarrow o} I_2$, which captures information flow but does not enforce physical realizability [23, 24, 25]; (iii) the Hamiltonian model of Chapter 3 and the cascade model of Chapter 5 are not equivalent [26], so a direct head-to-head comparison would require running each estimator on the other’s architecture; (iv) the EKF benchmark is a single representative trajectory, and full Monte-Carlo confidence intervals plus a sweep over $A_0, \gamma_s, \gamma_o, \lambda$ are needed before the quantitative claims generalize beyond the operating point reported here.

49. Perspectives

Three directions naturally extend this work. First, replacing the EKF by an unscented Kalman filter [39, 40] or a particle filter [41] would handle the cycle-slip regime where the EKF’s linearization bias and covariance shrinkage become significant. Second, the coherent-observer architecture sketched in the appendix could be turned into a hardware-realizable estimator by imposing physical realizability on the observer dynamics, replacing numerical differentiation by a smoothing stage, and validating on a coherent-feedback optical bench [13, 14, 15]. Third, comparing $J_\varphi(T)/\sigma_v^2$ to the quantum Fisher information of the output field [42, 17] would quantify how much phase information is captured by balanced homodyne, and how much is left on the table by restricting the measurement to a single quadrature.

References

- [1] M. Aspelmeyer, T. J. Kippenberg, and F. Marquardt, “Cavity optomechanics,” *Reviews of Modern Physics*, vol. 86, no. 4, pp. 1391–1452, 2014.
- [2] C. Weedbrook, S. Pirandola, R. García-Patrón, N. J. Cerf, T. C. Ralph, J. H. Shapiro, and S. Lloyd, “Gaussian quantum information,” *Reviews of Modern Physics*, vol. 84, no. 2, pp. 621–669, 2012.
- [3] H. M. Wiseman and G. J. Milburn, *Quantum Measurement and Control*. Cambridge, United Kingdom: Cambridge University Press, 2010.
- [4] C. W. Gardiner and P. Zoller, *Quantum Noise*. Berlin, Germany: Springer, 3 ed., 2004.
- [5] A. A. Clerk, M. H. Devoret, S. M. Girvin, F. Marquardt, and R. J. Schoelkopf, “Introduction to quantum noise, measurement, and amplification,” *Reviews of Modern Physics*, vol. 82, no. 2, pp. 1155–1208, 2010.
- [6] V. P. Belavkin, “Quantum stochastic calculus and quantum nonlinear filtering,” *Journal of Multivariate Analysis*, vol. 42, no. 2, pp. 171–201, 1992.
- [7] L. Bouten, R. van Handel, and M. R. James, “An introduction to quantum filtering,” *SIAM Journal on Control and Optimization*, vol. 46, no. 6, pp. 2199–2241, 2007.
- [8] H. M. Wiseman, “Adaptive phase measurements of optical modes: Going beyond the marginal q distribution,” *Physical Review Letters*, vol. 75, no. 25, pp. 4587–4590, 1995.
- [9] H. M. Wiseman and R. B. Killip, “Adaptive single-shot phase measurements: A semiclassical approach,” *Physical Review A*, vol. 57, no. 3, pp. 2169–2185, 1998.
- [10] T. A. Wheatley, D. W. Berry, H. Yonezawa, D. Nakane, H. Arao, D. T. Pope, T. C. Ralph, H. M. Wiseman, A. Furusawa, and E. H. Huntington, “Adaptive optical phase estimation using time-symmetric quantum smoothing,” *Physical Review Letters*, vol. 104, no. 9, p. 093601, 2010.
- [11] H. Yonezawa, D. Nakane, T. A. Wheatley, K. Iwasawa, S. Takeda, H. Arao, K. Ohki, K. Tsumura, D. W. Berry, T. C. Ralph, H. M. Wiseman, E. H. Huntington, and A. Furusawa, “Quantum-enhanced optical-phase tracking,” *Science*, vol. 337, no. 6101, pp. 1514–1517, 2012.
- [12] S. L. Vuglar and H. Amini, “Design of coherent quantum observers for linear quantum systems,” *New Journal of Physics*, vol. 16, no. 12, p. 125005, 2014.
- [13] Z. Miao, M. R. James, and I. R. Petersen, “Coherent observers for linear quantum stochastic systems,” *Automatica*, vol. 71, pp. 264–271, 2016.
- [14] J. E. Gough and N. H. Amini, “How to build a quantum coherent observer,” in *Proceedings of the 57th IEEE Conference on Decision and Control*, pp. 6597–6601, 2018.
- [15] S. Roy, I. R. Petersen, and E. H. Huntington, “Coherent-classical estimation for linear quantum systems,” *Automatica*, vol. 82, pp. 109–117, 2017.
- [16] H. Mabuchi, “Coherent-feedback quantum control with a dynamic compensator,” *Physical Review A*, vol. 78, no. 3, p. 032323, 2008.
- [17] M. Tsang, H. M. Wiseman, and C. M. Caves, “Fundamental quantum limit to waveform estimation,” *Physical Review Letters*, vol. 106, no. 9, p. 090401, 2011.

- [18] W. Wieczorek, S. G. Hofer, J. Hoelscher-Obermaier, R. Riedinger, K. Hammerer, and M. Aspelmeyer, “Optimal state estimation for cavity optomechanical systems,” *Physical Review Letters*, vol. 114, no. 22, p. 223601, 2015.
- [19] H. I. Nurdin and M. Guță, “Parameter estimation and system identification for continuously-observed quantum systems,” *Annual Reviews in Control*, vol. 54, pp. 295–304, 2022.
- [20] H. G. Clausen, P. Rouchon, and R. Wisniewski, “Online parameter estimation for continuously monitored quantum systems,” *IEEE Control Systems Letters*, 2024.
- [21] M. G. A. Paris, “Quantum estimation for quantum technology,” *International Journal of Quantum Information*, vol. 7, no. Supp 01, pp. 125–137, 2009.
- [22] V. Giovannetti, S. Lloyd, and L. Maccone, “Advances in quantum metrology,” *Nature Photonics*, vol. 5, no. 4, pp. 222–229, 2011.
- [23] M. R. James, H. I. Nurdin, and I. R. Petersen, “ h^∞ control of linear quantum stochastic systems,” *IEEE Transactions on Automatic Control*, vol. 53, no. 8, pp. 1787–1803, 2008.
- [24] S. L. Vuglar and I. R. Petersen, “Quantum noises, physical realizability and coherent quantum feedback control,” *IEEE Transactions on Automatic Control*, vol. 62, no. 2, pp. 998–1003, 2017.
- [25] G. Zhang and Z. Dong, “Linear quantum systems: A tutorial,” *Annual Reviews in Control*, vol. 54, pp. 274–294, 2022.
- [26] C. W. Gardiner, “Driving a quantum system with the output field from another driven quantum system,” *Physical Review Letters*, vol. 70, no. 15, pp. 2269–2272, 1993.
- [27] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *Transactions of the ASME—Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.
- [28] B. D. O. Anderson and J. B. Moore, *Optimal Filtering*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1979.
- [29] A. H. Jazwinski, *Stochastic Processes and Filtering Theory*. New York, NY, USA: Academic Press, 1970.
- [30] M. R. James, “Finite time observers and observability,” in *Proceedings of the 29th IEEE Conference on Decision and Control*, vol. 2, pp. 770–771, 1990.
- [31] C.-T. Chen, *Linear System Theory and Design*. New York, NY, USA: Oxford University Press, 3 ed., 1999.
- [32] C. W. Gardiner and M. J. Collett, “Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation,” *Physical Review A*, vol. 31, no. 6, pp. 3761–3774, 1985.
- [33] D. F. Walls and G. J. Milburn, *Quantum Optics*. Berlin, Germany: Springer, 2 ed., 2008.
- [34] A. J. Shaiju and I. R. Petersen, “On the physical realizability of general linear quantum stochastic differential equations with complex coefficients,” in *Proceedings of the 48th IEEE Conference on Decision and Control*, pp. 1422–1427, 2009.
- [35] C. L. Degen, F. Reinhard, and P. Cappellaro, “Quantum sensing,” *Reviews of Modern Physics*, vol. 89, no. 3, p. 035002, 2017.

-
- [36] J. Chen, B. B. Lane, S. Direkci, D. Ganapathy, X. Yin, N. Mavalvala, Y. Chen, and V. Sudhir, “Causal state estimation and the heisenberg uncertainty principle,” *Physical Review A*, vol. 110, no. 2, p. 022207, 2024.
 - [37] D. Simon, *Optimal State Estimation: Kalman, H_∞ , and Nonlinear Approaches*. Hoboken, NJ, USA: Wiley, 2006.
 - [38] D. W. Berry and H. M. Wiseman, “Adaptive quantum measurements of a continuously varying phase,” *Physical Review A*, vol. 65, no. 4, p. 043803, 2002.
 - [39] S. J. Julier and J. K. Uhlmann, “Unscented filtering and nonlinear estimation,” *Proceedings of the IEEE*, vol. 92, no. 3, pp. 401–422, 2004.
 - [40] E. A. Wan and R. Van Der Merwe, “The unscented Kalman filter for nonlinear estimation,” in *Proceedings of the IEEE Adaptive Systems for Signal Processing, Communications, and Control Symposium (AS-SPCC)*, pp. 153–158, 2000.
 - [41] A. Doucet, N. de Freitas, and N. Gordon, eds., *Sequential Monte Carlo Methods in Practice*. New York, NY, USA: Springer, 2001.
 - [42] S. L. Braunstein and C. M. Caves, “Statistical distance and the geometry of quantum states,” *Physical Review Letters*, vol. 72, no. 22, pp. 3439–3443, 1994.

Chapitre 8 – Appendix

50. Notation summary

For the convenience of the reader, the symbols used throughout the report are gathered in the Table. The same conventions are used in the main text and in this appendix. In particular, the hidden optical phase is consistently denoted by $\varphi(t)$ (never ϕ), the cavity linewidth by γ (never κ), and the homodyne measurement gain by κ_m . The phase-diffusion intensity is q_φ , and the relation $\sigma_\varphi^2 = q_\varphi$ is used in the chapter on Classical Extended Kalman Filtering for phase parameter estimation for compatibility with the discrete Euler formulation.

51. Balanced homodyne difference current

This appendix gives a detailed algebraic derivation of the balanced homodyne current used in the chapter on Homodyne Detection. The derivation follows the standard beam-splitter treatment of homodyne detection in quantum optics [3, 4, 5]. Let $\hat{a}_s(t)$ be the signal field carrying the hidden phase $\varphi(t)$, and let $\hat{b}_{\text{LO}}(t)$ be the local-oscillator field with controllable phase $\theta(t)$. With the beam-splitter convention used in the report,

$$\hat{c}(t) = \frac{\hat{a}_s(t) + \hat{b}_{\text{LO}}(t)}{\sqrt{2}}, \quad \hat{d}(t) = \frac{\hat{a}_s(t) - \hat{b}_{\text{LO}}(t)}{\sqrt{2}}. \quad (177)$$

The two photodiode currents are proportional to the photon-number operators $\hat{c}^\dagger \hat{c}$ and $\hat{d}^\dagger \hat{d}$. The balanced (difference) current is therefore

$$\hat{I}_{\text{diff}}(t) \propto \hat{c}^\dagger(t) \hat{c}(t) - \hat{d}^\dagger(t) \hat{d}(t). \quad (178)$$

Expansion of $\hat{c}^\dagger \hat{c}$ and $\hat{d}^\dagger \hat{d}$. Substituting (177) and using $(\hat{a} + \hat{b})^\dagger = \hat{a}^\dagger + \hat{b}^\dagger$, one obtains

$$\begin{aligned} \hat{c}^\dagger \hat{c} &= \frac{1}{2} (\hat{a}_s^\dagger + \hat{b}_{\text{LO}}^\dagger) (\hat{a}_s + \hat{b}_{\text{LO}}) \\ &= \frac{1}{2} (\hat{a}_s^\dagger \hat{a}_s + \hat{a}_s^\dagger \hat{b}_{\text{LO}} + \hat{b}_{\text{LO}}^\dagger \hat{a}_s + \hat{b}_{\text{LO}}^\dagger \hat{b}_{\text{LO}}), \end{aligned} \quad (179)$$

$$\begin{aligned} \hat{d}^\dagger \hat{d} &= \frac{1}{2} (\hat{a}_s^\dagger - \hat{b}_{\text{LO}}^\dagger) (\hat{a}_s - \hat{b}_{\text{LO}}) \\ &= \frac{1}{2} (\hat{a}_s^\dagger \hat{a}_s - \hat{a}_s^\dagger \hat{b}_{\text{LO}} - \hat{b}_{\text{LO}}^\dagger \hat{a}_s + \hat{b}_{\text{LO}}^\dagger \hat{b}_{\text{LO}}). \end{aligned} \quad (180)$$

Subtracting (180) from (179), the direct signal intensity $\hat{a}_s^\dagger \hat{a}_s$ and the direct local-oscillator intensity $\hat{b}_{\text{LO}}^\dagger \hat{b}_{\text{LO}}$ cancel:

$$\boxed{\hat{c}^\dagger \hat{c} - \hat{d}^\dagger \hat{d} = \hat{a}_s^\dagger \hat{b}_{\text{LO}} + \hat{b}_{\text{LO}}^\dagger \hat{a}_s.} \quad (181)$$

Only the interference term between the signal and the local oscillator survives.

Classical-limit substitution. Replacing the operators by coherent amplitudes

$$a_s(t) = A_s e^{i\varphi(t)}, \quad b_{\text{LO}}(t) = B_{\text{LO}} e^{i\theta(t)},$$

and using $(e^{i\alpha})^* = e^{-i\alpha}$,

$$\begin{aligned} a_s^* b_{\text{LO}} + b_{\text{LO}}^* a_s &= A_s B_{\text{LO}} \left[e^{-i\varphi(t)} e^{i\theta(t)} + e^{-i\theta(t)} e^{i\varphi(t)} \right] \\ &= A_s B_{\text{LO}} \left[e^{i(\theta-\varphi)} + e^{i(\varphi-\theta)} \right] \\ &= 2A_s B_{\text{LO}} \cos(\varphi(t) - \theta(t)), \end{aligned} \quad (182)$$

where the last equality uses Euler's formula $\cos x = (e^{ix} + e^{-ix})/2$.

Classical homodyne current. Hence, with the convention used in this report,

$$I_{\text{diff}}(t) \propto 2A_s B_{\text{LO}} \cos(\varphi(t) - \theta(t)). \quad (183)$$

Other beam-splitter conventions may produce a sine instead of a cosine; this only corresponds to a shift of the local-oscillator phase by $\pi/2$ and does not change the physics.

Stochastic measurement model. Including detector calibration and shot noise, the measurement record is modeled as the stochastic differential equation

$$dY(t) = K_h \cos(\varphi(t) - \theta(t)) dt + \sigma_h dV(t), \quad (184)$$

where $K_h > 0$ is the calibrated homodyne gain (which absorbs $2A_s B_{\text{LO}}$ and the photodetector responsivity), $\sigma_h > 0$ is the measurement-noise amplitude, and $V(t)$ is a standard Wiener process representing shot noise.

52. Scalar Kalman model for OU phase estimation

The scalar phase model used in the first estimation example (chapter on the Classical Observer for Phase Estimation via Homodyne Detection) is

$$d\varphi(t) = -\lambda\varphi(t) dt + \sqrt{q_\varphi} dW_\varphi(t), \quad (185)$$

where $\lambda > 0$ is the relaxation rate, $q_\varphi > 0$ is the phase-diffusion intensity, and $W_\varphi(t)$ is a standard Wiener process. After homodyne linearization and normalization, the measurement is

$$dy(t) = \varphi(t) dt + \frac{1}{2|\alpha|} dV(t), \quad (186)$$

where $|\alpha|$ is the local-oscillator amplitude and $V(t)$ is a standard Wiener process independent of $W_\varphi(t)$.

Identification of the canonical state-space matrices Comparing (185)–(186) with the canonical scalar linear–Gaussian system

$$dx = Ax dt + G dW_\varphi, \quad dy = Cx dt + D dV,$$

yields

$$A = -\lambda, \quad G = \sqrt{q_\varphi}, \quad C = 1, \quad D = \frac{1}{2|\alpha|}.$$

The general matrix Riccati equation for the Kalman–Bucy error covariance reads

$$\dot{P}(t) = AP(t) + P(t)A^\top + GG^\top - P(t)C^\top(DD^\top)^{-1}CP(t).$$

In the scalar case, P, A, C, D, G are real numbers, so $A^\top = A$, $C^\top = C$, and $GG^\top = G^2 = q_\varphi$, $DD^\top = D^2 = 1/(4|\alpha|^2)$. The quadratic correction term is

$$PC(DD^\top)^{-1}CP = P \cdot 1 \cdot 4|\alpha|^2 \cdot 1 \cdot P = 4|\alpha|^2 P^2.$$

Combining,

$$\boxed{\dot{P}(t) = -2\lambda P(t) + q_\varphi - 4|\alpha|^2 P(t)^2.} \quad (187)$$

The corresponding scalar Kalman gain is

$$K(t) = \frac{P(t)C^\top}{D^2} = 4|\alpha|^2 P(t). \quad (188)$$

Steady-state solution. Setting $\dot{P} = 0$ in (187) yields

$$4|\alpha|^2 P_\infty^2 + 2\lambda P_\infty - q_\varphi = 0.$$

This is a quadratic equation in P_∞ . Its discriminant is $\Delta = (2\lambda)^2 + 4 \cdot 4|\alpha|^2 q_\varphi = 4(\lambda^2 + 4|\alpha|^2 q_\varphi)$, so the two roots are

$$P_\infty^\pm = \frac{-2\lambda \pm 2\sqrt{\lambda^2 + 4|\alpha|^2 q_\varphi}}{8|\alpha|^2} = \frac{-\lambda \pm \sqrt{\lambda^2 + 4|\alpha|^2 q_\varphi}}{4|\alpha|^2}.$$

Only the positive root is physically admissible (since $P_\infty \geq 0$):

$$P_\infty = \frac{-\lambda + \sqrt{\lambda^2 + 4|\alpha|^2 q_\varphi}}{4|\alpha|^2}. \quad (189)$$

Equivalently, multiplying numerator and denominator by the conjugate, one recovers the form given in the main text,

$$P_\infty = \frac{1}{4|\alpha|^2} \left(-\lambda + \sqrt{\lambda^2 + 4|\alpha|^2 q_\varphi} \right) = R_h \left(-\lambda + \sqrt{\lambda^2 + \frac{q_\varphi}{R_h}} \right), \quad R_h = \frac{1}{4|\alpha|^2}.$$

With $t_k = k\Delta t$, denoting $P_k \simeq P(t_k)$, the explicit Euler scheme applied to (187) gives

$$P_{k+1} = P_k + \Delta t \left(-2\lambda P_k + q_\varphi - 4|\alpha|^2 P_k^2 \right). \quad (190)$$

The estimator update follows the standard Kalman–Bucy structure

$$d\hat{\varphi}(t) = -\lambda \hat{\varphi}(t) dt + K(t) \left(dy(t) - \hat{\varphi}(t) dt \right),$$

which, discretized on the same grid, becomes

$$\hat{\varphi}_{k+1} = \hat{\varphi}_k + \Delta t \left[-\lambda \hat{\varphi}_k + K_{k+1} \left(y_k - \hat{\varphi}_k \right) \right]. \quad (191)$$

53. Cavity Langevin equation and quadrature model

The single-mode cavity model used in the report follows the standard input–output formalism for open quantum systems [4, 3, 25]. For a cavity mode $a(t)$, detuning Δ , linewidth γ , and input field $a_{\text{in}}(t)$, the Markov approximation gives

$$\dot{a}(t) = -\left(\frac{\gamma}{2} + i\Delta\right) a(t) + \sqrt{\gamma} a_{\text{in}}(t). \quad (192)$$

The factor $\sqrt{\gamma}$ appears because γ is a decay rate, while $a_{\text{in}}(t)$ has the normalization of a quantum white-noise field ($[a_{\text{in}}(t), a_{\text{in}}^\dagger(t')] = \delta(t - t')$). With the convention used here, the corresponding input–output relation is

$$a_{\text{out}}(t) = a_{\text{in}}(t) - \sqrt{\gamma} a(t). \quad (193)$$

For the linearized phase-sensitive cavity model, we use

$$\dot{a}(t) = \left(-\frac{\gamma}{2} - i\omega\right) a(t) + ig_\varphi \varphi(t) + \sqrt{\gamma} a_{\text{in}}(t), \quad (194)$$

where ω is the rotating-frame detuning and g_φ is the phase-to-field coupling (linearized around the steady coherent operating point, cf. the chapter on the Optical cavity formulation as a phase-estimation problem). Defining the quadratures (consistent with the convention of the chapter on the Optical cavity formulation as a phase-estimation problem, in which $[\hat{q}, \hat{p}] = 2i$):

$$q(t) = a(t) + a^\dagger(t), \quad p(t) = \frac{a(t) - a^\dagger(t)}{i}. \quad (195)$$

Equivalently, $a(t) = \frac{1}{2}(q(t) + ip(t))$, so that $\dot{a} = \frac{1}{2}(\dot{q} + i\dot{p})$. Taking the real and imaginary parts of (194) requires expanding each term: (i) Drift: $\left(-\frac{\gamma}{2} - i\omega\right) a = \left(-\frac{\gamma}{2} - i\omega\right) \cdot \frac{1}{2}(q + ip) = \frac{1}{2}\left[-\frac{\gamma}{2}q + \omega p\right] + \frac{i}{2}\left[-\omega q - \frac{\gamma}{2}p\right]$. (ii) Phase forcing: $ig_\varphi \varphi(t)$ is purely imaginary, hence contributes only to the p -equation. (iii) Input noise: writing $q_{\text{in}} = a_{\text{in}} + a_{\text{in}}^\dagger$ and $p_{\text{in}} = (a_{\text{in}} - a_{\text{in}}^\dagger)/i$, one has $a_{\text{in}} = \frac{1}{2}(q_{\text{in}} + ip_{\text{in}})$. Identifying $\dot{q}$ and $\dot{p}$ by matching real and imaginary parts gives

$$\dot{q}(t) = -\frac{\gamma}{2}q(t) + \omega p(t) + \sqrt{\gamma} q_{\text{in}}(t), \quad (196)$$

$$\dot{p}(t) = -\omega q(t) - \frac{\gamma}{2}p(t) + g_\varphi \varphi(t) + \sqrt{\gamma} p_{\text{in}}(t). \quad (197)$$

The hidden phase $\varphi(t)$ appears only in the p -equation: this is the structural origin of the indirect phase-estimation problem. Stacking $x(t) = [q(t), p(t)]^\top$, the centered quadrature model reads

$$\dot{x}(t) = Ax(t) + E\varphi(t) + G\xi(t), \quad A = \begin{bmatrix} -\gamma/2 & \omega \\ -\omega & -\gamma/2 \end{bmatrix}, \quad E = \begin{bmatrix} 0 \\ g_\varphi \end{bmatrix}. \quad (198)$$

If $\varphi(t)$ follows the Ornstein–Uhlenbeck dynamics $d\varphi = -\lambda\varphi dt + \sqrt{q_\varphi} dW_\varphi$, the augmented state $x_a(t) = [q(t), p(t), \varphi(t)]^\top$ has drift matrix

$$A_a = \begin{bmatrix} -\gamma/2 & \omega & 0 \\ -\omega & -\gamma/2 & g_\varphi \\ 0 & 0 & -\lambda \end{bmatrix}. \quad (199)$$

The block-triangular structure (zero entries in the third row except $-\lambda$) reflects the fact that the phase drives the cavity but is not driven by it.

54. Proof of the steady-state phase-variance decomposition

We give the detailed proof of the decomposition used in the chapter on Cavity Phase Estimation: Observability Limits and Regime-Dependent Preprocessing Trade-Offs. At steady state, the Kalman–Bucy covariance P_∞ satisfies

$$A_a P_\infty + P_\infty A_a^\top + Q - P_\infty C_a^\top R^{-1} C_a P_\infty = 0, \quad (200)$$

where $Q = G_a G_a^\top$ is the augmented process-noise covariance and $R = \sigma_v^2$ is the measurement-noise variance.

Resonant case $\omega = 0$. Take

$$A_a = \begin{bmatrix} -\gamma/2 & 0 & 0 \\ 0 & -\gamma/2 & g_\varphi \\ 0 & 0 & -\lambda \end{bmatrix}, \quad C_a = \sqrt{\kappa_m} \begin{bmatrix} \cos \theta & \sin \theta & 0 \end{bmatrix},$$

and write the symmetric covariance as

$$P_\infty = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{12} & p_{22} & p_{23} \\ p_{13} & p_{23} & p_{33} \end{bmatrix}.$$

(3,3) entry of the linear part. The third row of A_a is $[0, 0, -\lambda]$, so $(A_a P_\infty)_{33} = -\lambda p_{33}$. Symmetrically, $(P_\infty A_a^\top)_{33} = -\lambda p_{33}$. The contribution of Q to position (3,3) is $Q_{33} = q_\varphi$ (since the (3,3) component of the augmented process-noise corresponds to the OU phase diffusion).

(3,3) entry of the quadratic part. We compute $\left[P_\infty C_a^\top R^{-1} C_a P_\infty \right]_{33}$. Note that $(C_a P_\infty)$ is a row vector whose third component is

$$(C_a P_\infty)_3 = \sqrt{\kappa_m} (\cos \theta p_{13} + \sin \theta p_{23} + 0 \cdot p_{33}) = \sqrt{\kappa_m} p_{\theta\varphi},$$

where we have set $p_{\theta\varphi} \equiv \cos \theta p_{13} + \sin \theta p_{23}$. Hence

$$\left[P_\infty C_a^\top R^{-1} C_a P_\infty \right]_{33} = (C_a P_\infty)_3 \cdot R^{-1} \cdot (C_a P_\infty)_3^\top = \frac{\kappa_m p_{\theta\varphi}^2}{\sigma_v^2}.$$

Assembling the scalar equation. Summing the three contributions, the (3,3) component of (200) reads

$$-2\lambda p_{33} + q_\varphi - \frac{\kappa_m p_{\theta\varphi}^2}{\sigma_v^2} = 0.$$

Solving for $p_{33} = [P_\infty]_{\varphi\varphi}$,

$$\boxed{[P_\infty]_{\varphi\varphi} = \frac{q_\varphi}{2\lambda} - \frac{\kappa_m}{2\lambda\sigma_v^2} (p_{13} \cos \theta + p_{23} \sin \theta)^2.} \quad (201)$$

The first term $q_\varphi/(2\lambda)$ is the open-loop stationary variance of the OU phase (see (185)). The second term is a non-positive correction induced by measurement information: it is proportional to κ_m/σ_v^2 (signal-to-noise of the homodyne channel) and to the squared closed-loop covariance $p_{\theta\varphi}^2$ between the measured quadrature direction and the hidden phase. Because $p_{\theta\varphi}$ is determined implicitly by the full Riccati system, the steady-state phase variance is not a function of the directional visibility J_φ alone.

55. Closed form of the directional phase visibility

The directional phase visibility introduced in the chapter on Cavity Phase Estimation: Observability Limits and Regime-Dependent Preprocessing Trade-Offs is defined by

$$J_\varphi(\infty) = \int_0^\infty \|C_a e^{A_a t} e_\varphi\|^2 dt, \quad e_\varphi = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (202)$$

Resonant decoupled drift matrix. In the resonant case $\omega = 0$, with the shorthand $\mu = \gamma/2$,

$$A_a = \begin{bmatrix} -\mu & 0 & 0 \\ 0 & -\mu & g_\varphi \\ 0 & 0 & -\lambda \end{bmatrix}.$$

The matrix A_a is block-triangular with 2×2 upper block $\text{diag}(-\mu, -\mu)$ and 1×1 lower block $(-\lambda)$. Its exponential is computed component-wise.

Solving $\dot{x} = A_a x$ with $x(0) = e_\varphi$. The third equation is $\dot{x}_3 = -\lambda x_3$ with $x_3(0) = 1$, so $x_3(t) = e^{-\lambda t}$. The first equation is $\dot{x}_1 = -\mu x_1$ with $x_1(0) = 0$, so $x_1(t) = 0$. The second equation is $\dot{x}_2 = -\mu x_2 + g_\varphi x_3 = -\mu x_2 + g_\varphi e^{-\lambda t}$, with $x_2(0) = 0$. Using the integrating factor $e^{\mu t}$,

$$\frac{d}{dt} (e^{\mu t} x_2(t)) = g_\varphi e^{(\mu-\lambda)t}.$$

Integrating from 0 to t (with $\mu \neq \lambda$),

$$e^{\mu t} x_2(t) = \frac{g_\varphi}{\mu - \lambda} (e^{(\mu-\lambda)t} - 1),$$

hence

$$x_2(t) = \frac{g_\varphi}{\mu - \lambda} (e^{-\lambda t} - e^{-\mu t}).$$

The case $\mu = \lambda$ follows by continuity: $x_2(t) = g_\varphi t e^{-\lambda t}$. Combining,

$$e^{A_a t} e_\varphi = \begin{bmatrix} 0 \\ \frac{g_\varphi}{\mu - \lambda} (e^{-\lambda t} - e^{-\mu t}) \\ e^{-\lambda t} \end{bmatrix}. \quad (203)$$

Projection onto the homodyne row. Applying $C_a = \sqrt{\kappa_m} [\cos \theta, \sin \theta, 0]$ selects only the second component:

$$C_a e^{A_a t} e_\varphi = \sqrt{\kappa_m} \sin \theta \cdot \frac{g_\varphi}{\mu - \lambda} (e^{-\lambda t} - e^{-\mu t}). \quad (204)$$

Squaring and integrating,

$$\begin{aligned} J_\varphi(\infty) &= \frac{\kappa_m g_\varphi^2 \sin^2 \theta}{(\mu - \lambda)^2} \int_0^\infty (e^{-\lambda t} - e^{-\mu t})^2 dt \\ &= \frac{\kappa_m g_\varphi^2 \sin^2 \theta}{(\mu - \lambda)^2} \int_0^\infty (e^{-2\lambda t} - 2e^{-(\lambda+\mu)t} + e^{-2\mu t}) dt \\ &= \frac{\kappa_m g_\varphi^2 \sin^2 \theta}{(\mu - \lambda)^2} \left[\frac{1}{2\lambda} - \frac{2}{\lambda + \mu} + \frac{1}{2\mu} \right]. \end{aligned} \quad (205)$$

The bracket simplifies as follows. Putting all three fractions over the common denominator $2\lambda\mu(\lambda + \mu)$,

$$\begin{aligned} \frac{1}{2\lambda} - \frac{2}{\lambda + \mu} + \frac{1}{2\mu} &= \frac{\mu(\lambda + \mu) - 4\lambda\mu + \lambda(\lambda + \mu)}{2\lambda\mu(\lambda + \mu)} \\ &= \frac{\lambda^2 + 2\lambda\mu + \mu^2 - 4\lambda\mu}{2\lambda\mu(\lambda + \mu)} \\ &= \frac{(\mu - \lambda)^2}{2\lambda\mu(\lambda + \mu)}. \end{aligned}$$

The factor $(\mu - \lambda)^2$ cancels exactly with the prefactor of (205), removing the apparent singularity at $\mu = \lambda$:

$$J_\varphi(\infty) = \frac{g_\varphi^2 \kappa_m \sin^2 \theta}{2\lambda (\gamma/2) (\lambda + \gamma/2)}. \quad (206)$$

This expression shows that $J_\varphi(\infty)$ captures the structural visibility of the phase direction through the homodyne channel. It does not, by itself, determine the full stationary Kalman phase variance, because the latter also depends on the Riccati covariance correction in (201).

Continuity at $\mu = \lambda$. A direct computation in the case $\mu = \lambda$ (using $x_2(t) = g_\varphi t e^{-\lambda t}$) gives

$$J_\varphi(\infty) \Big|_{\mu=\lambda} = \kappa_m g_\varphi^2 \sin^2 \theta \int_0^\infty t^2 e^{-2\lambda t} dt = \kappa_m g_\varphi^2 \sin^2 \theta \cdot \frac{1}{4\lambda^3},$$

which equals the limit of (206) when $\mu \rightarrow \lambda$, confirming continuity.

56. EKF Jacobians used in the chapter on Classical Extended Kalman Filtering

For the EKF defined in the chapter on Classical Extended Kalman Filtering for phase parameter estimation, the Jacobian of the discrete state map f is evaluated at

$$\widehat{X}_{k-1|k-1} = \begin{bmatrix} \widehat{q}_{s,k-1} \\ \widehat{p}_{s,k-1} \\ \widehat{q}_{o,k-1} \\ \widehat{p}_{o,k-1} \\ \widehat{\varphi}_{k-1|k-1} \end{bmatrix}.$$

Differentiating the discrete state map block by block yields, with γ_s, γ_o for the sensing and auxiliary linewidths (consistent with the main text),

$$F_k = \begin{bmatrix} 1 - \frac{\gamma_s dt}{2} & \omega_s dt & 0 & 0 & -dt\sqrt{2\gamma_s} A_0 \sin \widehat{\varphi}_{k-1|k-1} \\ -\omega_s dt & 1 - \frac{\gamma_s dt}{2} & 0 & 0 & dt\sqrt{2\gamma_s} A_0 \cos \widehat{\varphi}_{k-1|k-1} \\ -dt\sqrt{\gamma_s \gamma_o} & 0 & 1 - \frac{\gamma_o dt}{2} & \omega_o dt & -dt\sqrt{2\gamma_o} A_0 \sin \widehat{\varphi}_{k-1|k-1} \\ 0 & -dt\sqrt{\gamma_s \gamma_o} & -\omega_o dt & 1 - \frac{\gamma_o dt}{2} & dt\sqrt{2\gamma_o} A_0 \cos \widehat{\varphi}_{k-1|k-1} \\ 0 & 0 & 0 & 0 & 1 - \lambda dt \end{bmatrix}. \quad (207)$$

Origin of the off-diagonal $-dt\sqrt{\gamma_s \gamma_o}$ terms. These terms come from the cascade input-output relation (193) applied to the auxiliary cavity: the auxiliary state receives, as part of its input, the output field of the sensing cavity, which contains the term $-\sqrt{\gamma_s} a_s$. When projected onto the auxiliary quadratures and discretized at first order in dt , this produces the $-dt\sqrt{\gamma_s \gamma_o}$ coefficient in front of $(q_{s,k-1}, p_{s,k-1})$. It encodes the unidirectional flow of information from the sensing cavity to the auxiliary cavity.

Measurement Jacobian $H_k = \partial h / \partial X$. At time step k , the homodyne angle θ_k is treated as known when differentiating the measurement function $h(X, \theta_k)$. Differentiating the explicit form of h with respect to each component of X gives

$$H_k = \begin{bmatrix} -\sqrt{\gamma_s} \cos \theta_k & -\sqrt{\gamma_s} \sin \theta_k & -\sqrt{\gamma_o} \cos \theta_k & -\sqrt{\gamma_o} \sin \theta_k & -\sqrt{2} A_0 \sin(\hat{\varphi}_{k|k-1} - \theta_k) \end{bmatrix}. \quad (208)$$

The trigonometric identity used to obtain the last component is $\cos \theta_k (-\sin \hat{\varphi}_{k|k-1}) + \sin \theta_k (\cos \hat{\varphi}_{k|k-1}) = \sin(\theta_k - \hat{\varphi}_{k|k-1}) = -\sin(\hat{\varphi}_{k|k-1} - \theta_k)$, multiplied by $\sqrt{2} A_0$. A simple block-diagonal process-noise covariance used in the simulations is

$$Q = \text{blkdiag}(\gamma_s dt I_2, \gamma_o dt I_2, \sigma_\varphi^2 dt), \quad (209)$$

with measurement-noise variance R . The first two blocks are the vacuum-driven cavity quadrature noises with intensity γ_s, γ_o ; the last block is the OU phase diffusion with intensity $\sigma_\varphi^2 = q_\varphi$.